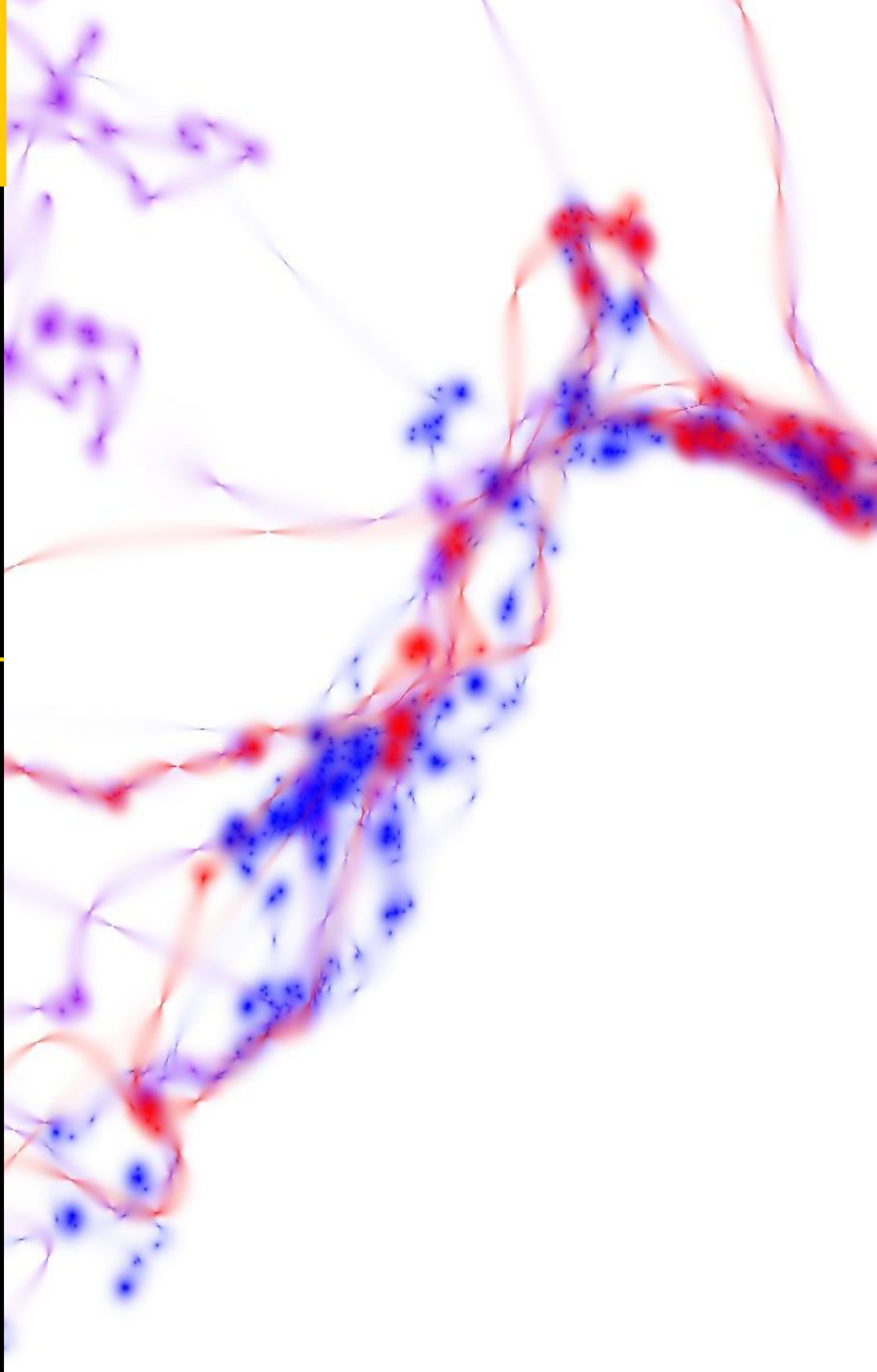
Introduction to Continuous-Time Movement Modeling

Continuous-Time Animal Movement Modeling

Christen H. Fleming , Inês Silva, Erika Y. Lin
University of Central Florida
christen.fleming@ucf.edu
i.simoes-silva@hzdr.de
erika.lin@ucf.edu



Fleming

Overview

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1) Introduction
2) The problem: Underestimating space use
3) The solution
4) Continuous-time vs Discrete-time
5) Accounting for non-independence
6) `ctmm`

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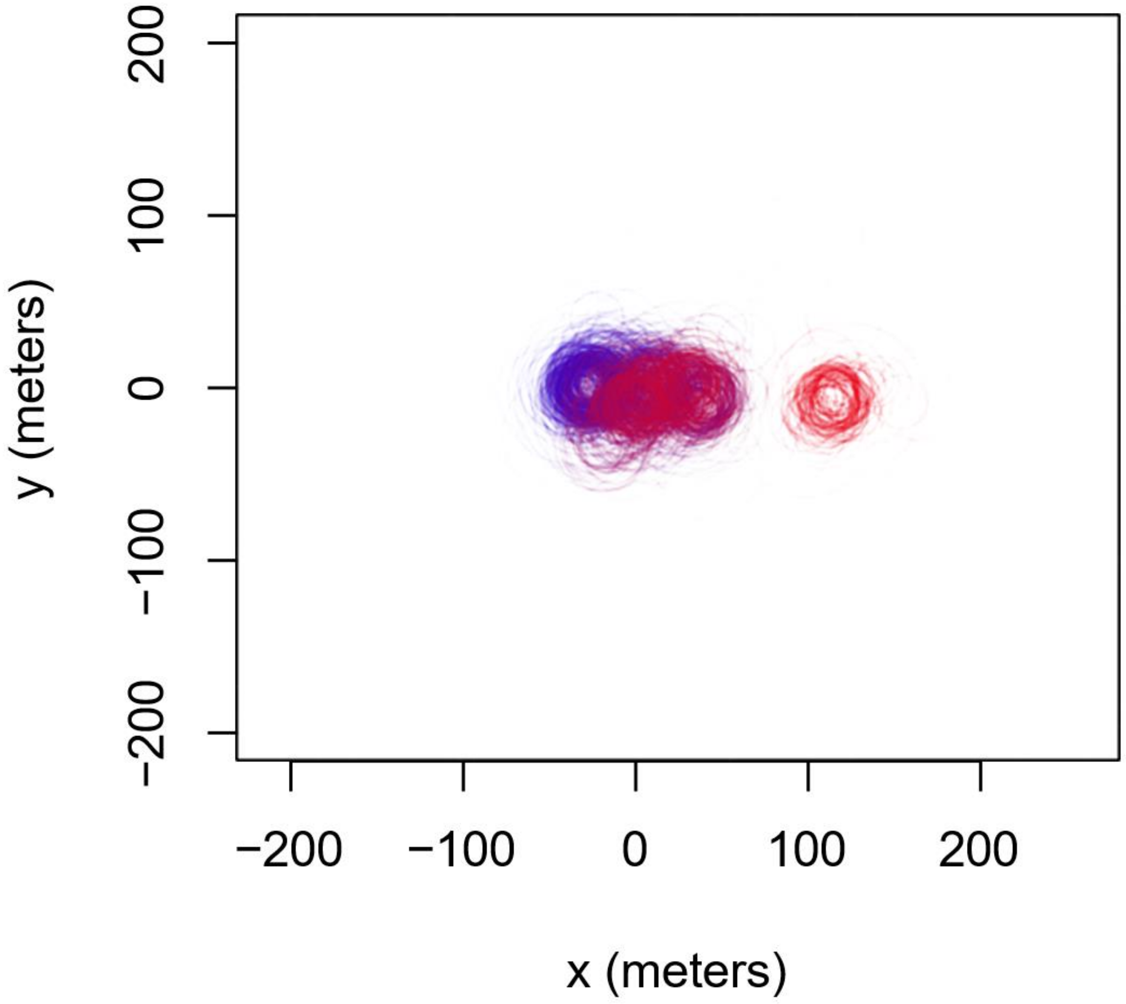
Introduction

- Animal tracking data are complex

Wood turtle
(Glyptemys insculpta)



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Animal tracking data analysis goals in this workshop:

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- Account for autocorrelation

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- Account for autocorrelation (stochastic process models)

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- Don’t be satisfied with point estimates

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- Reducing bias and error as much as possible (MVU, BLUE, debiased estimators)

Autocorrelation

- What is autocorrelation?
- Why is it important to consider when modeling animal movement?

Autocorrelation

First law of Geography:

> (...) Everything is related to everything else, but near things are more related than distant things.
>
> Tobler (1970)

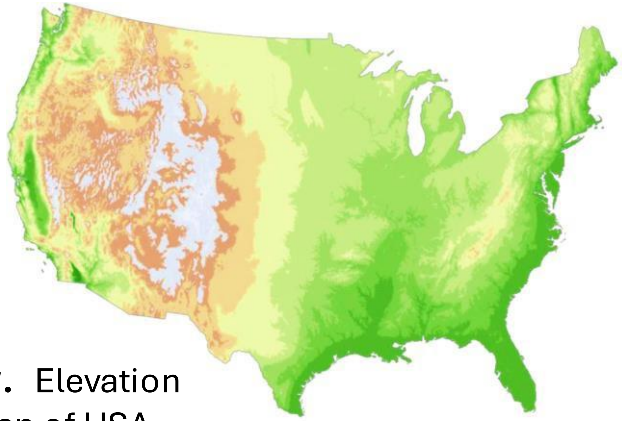


Fig. Elevation map of USA.

Autocorrelation

- **Autocorrelation**: correlation of values in a variable with itself
  - Similarity between variables as a function of distance (*spatial*), time (*temporal*), relatedness (*phylogenetic*), etc.

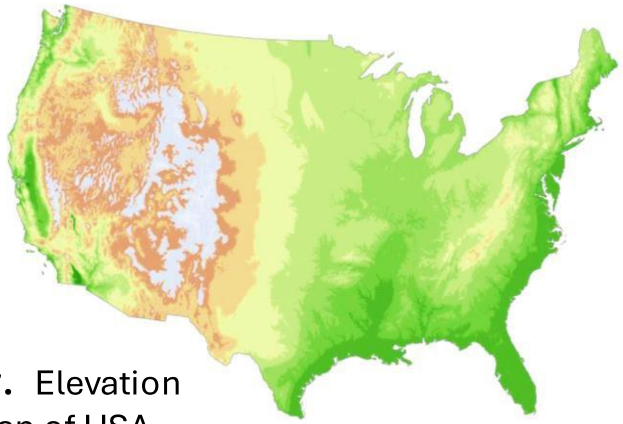


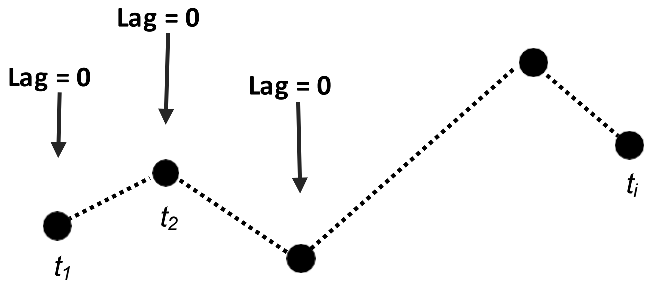
Fig. Elevation map of USA.

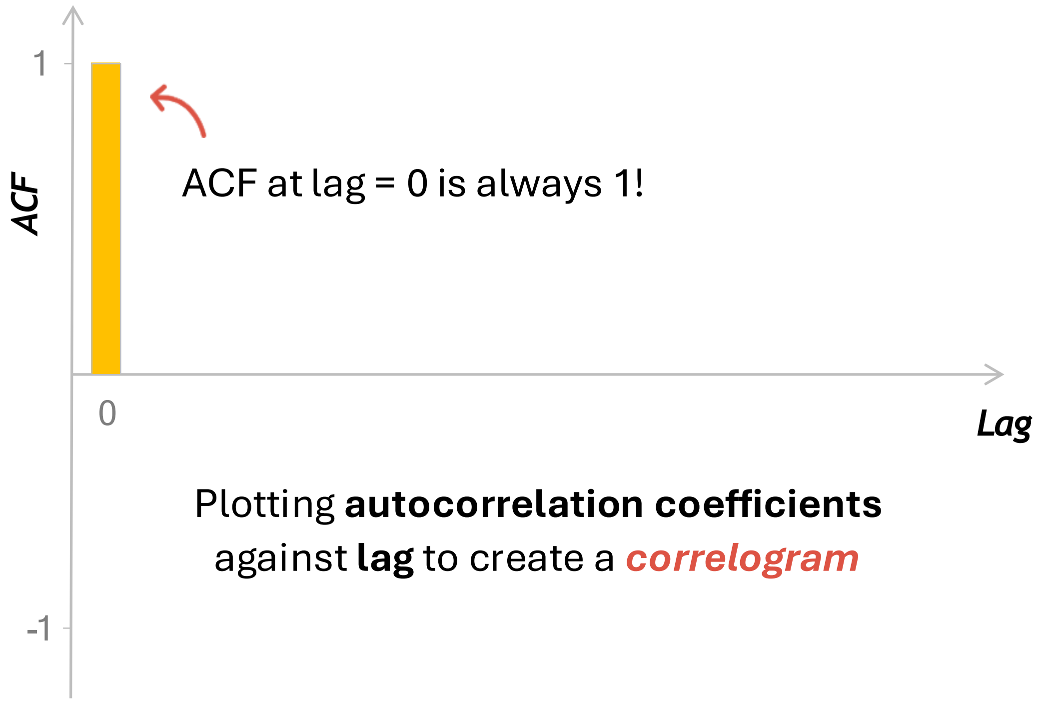
If features were not autocorrelated

How can we measure and visualize autocorrelation?

- The **autocorrelation function** (ACF) is a valuable tool for **detecting** autocorrelation in *time series data*
  - This is usually visualized through a correlogram
  - Ranges from -1 to 1

$$\mathrm{ACF}_{lag} = COV[x_t, x_{t+lag}]/Var[X_t]$$





Plotting **autocorrelation coefficients** against **lag** to create a ***correlogram***

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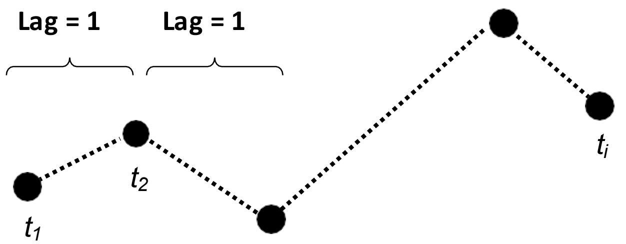
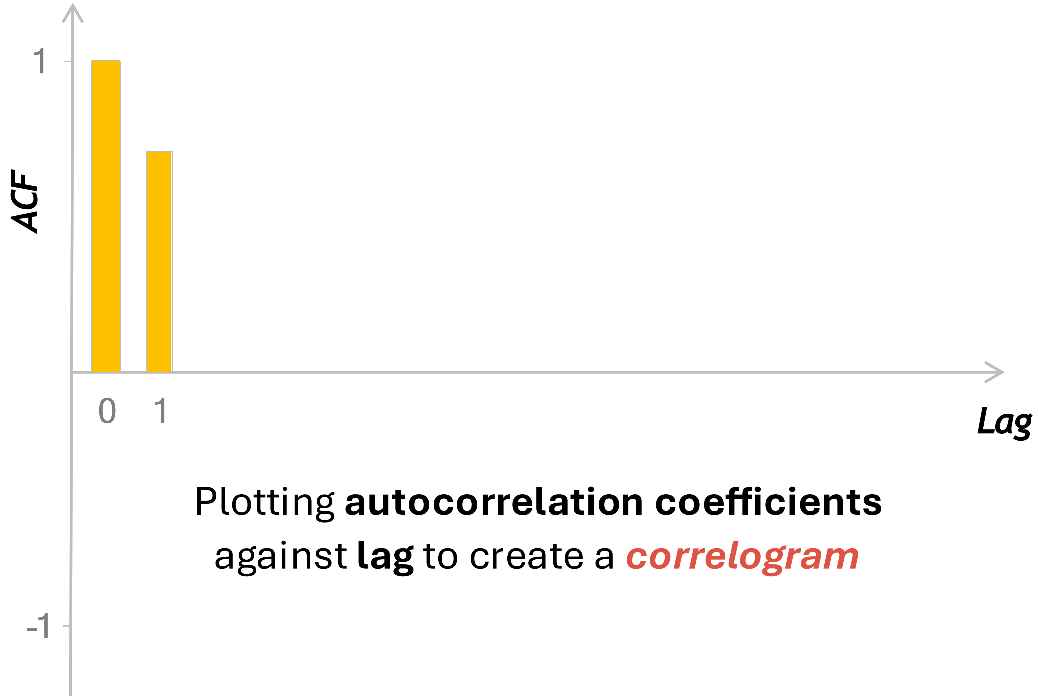
$$\mathbf{ACF_1} = \boldsymbol{COV[x_t, x_{t+1}]/Var[X_t]}$$





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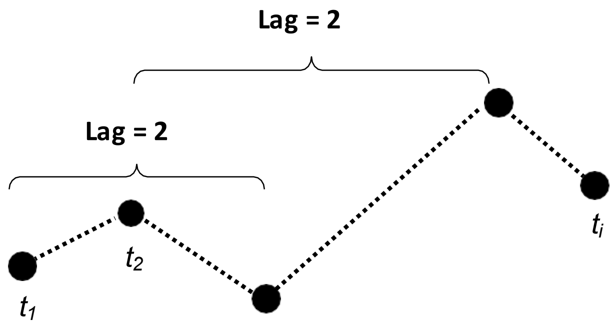
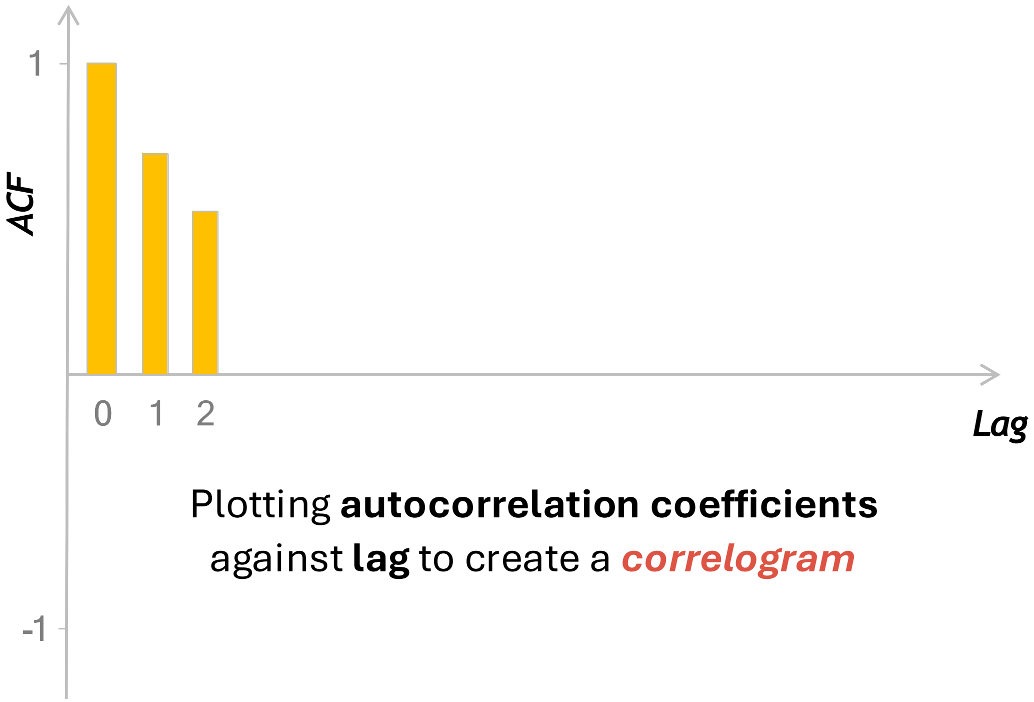
$$\mathrm{ACF}_2 = COV[x_t, x_{t+2}]/Var[X_t]$$

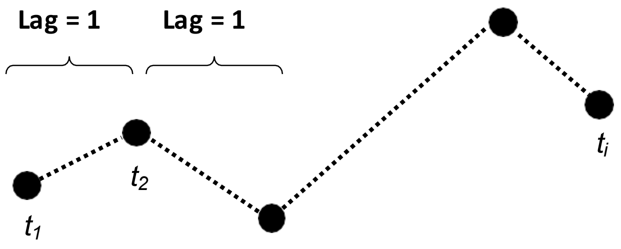


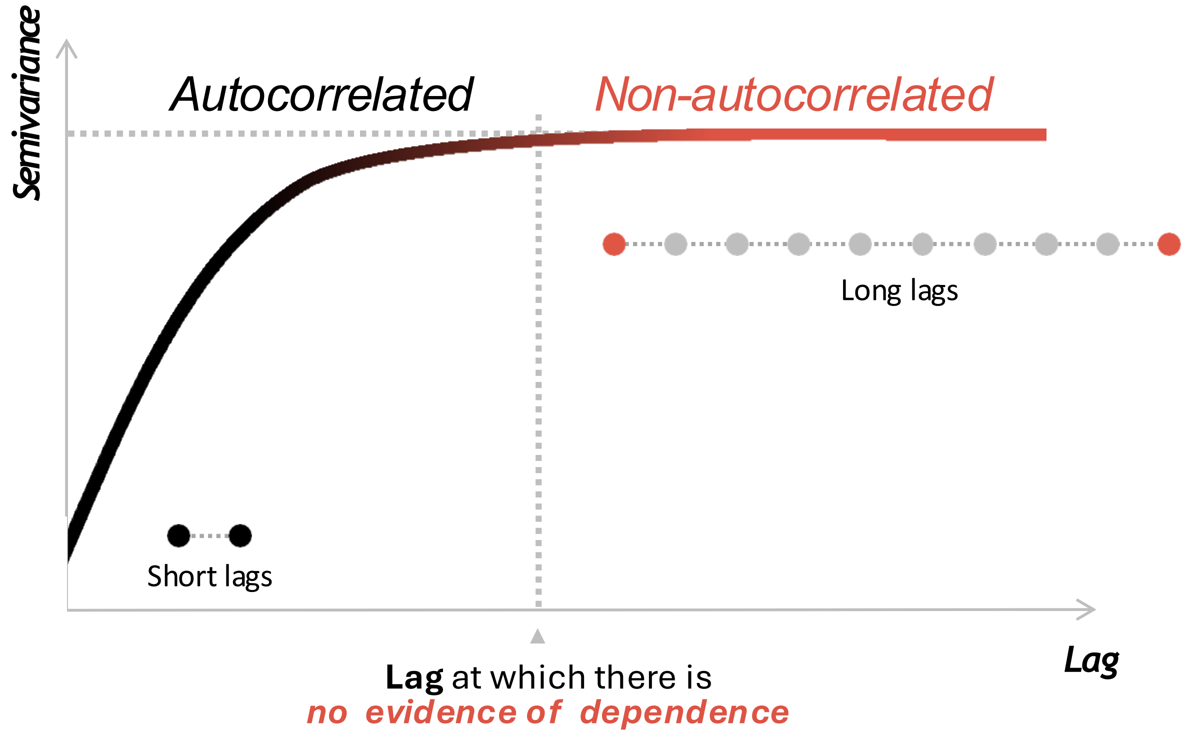


Plotting **autocorrelation coefficients** against **lag** to create a ***correlogram***

How can we measure and visualize autocorrelation?

Semivariograms (or variograms) are used to **visualize** autocorrelation. We plot lags on the x-axis for all pairs of observation against their semivariance (mean square distance between any 2 locations with a given lag between) on the y-axis.





The *more similar* the pairs of locations per lag, the *lower the semivariance* for that lag.

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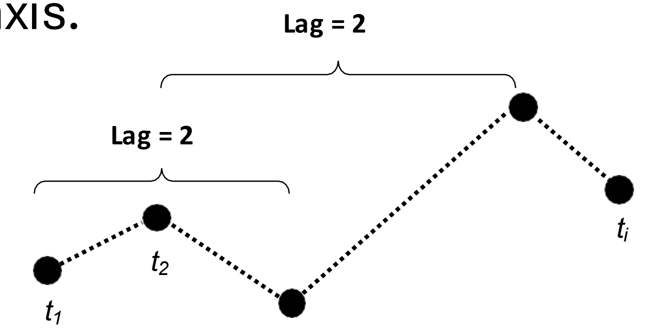

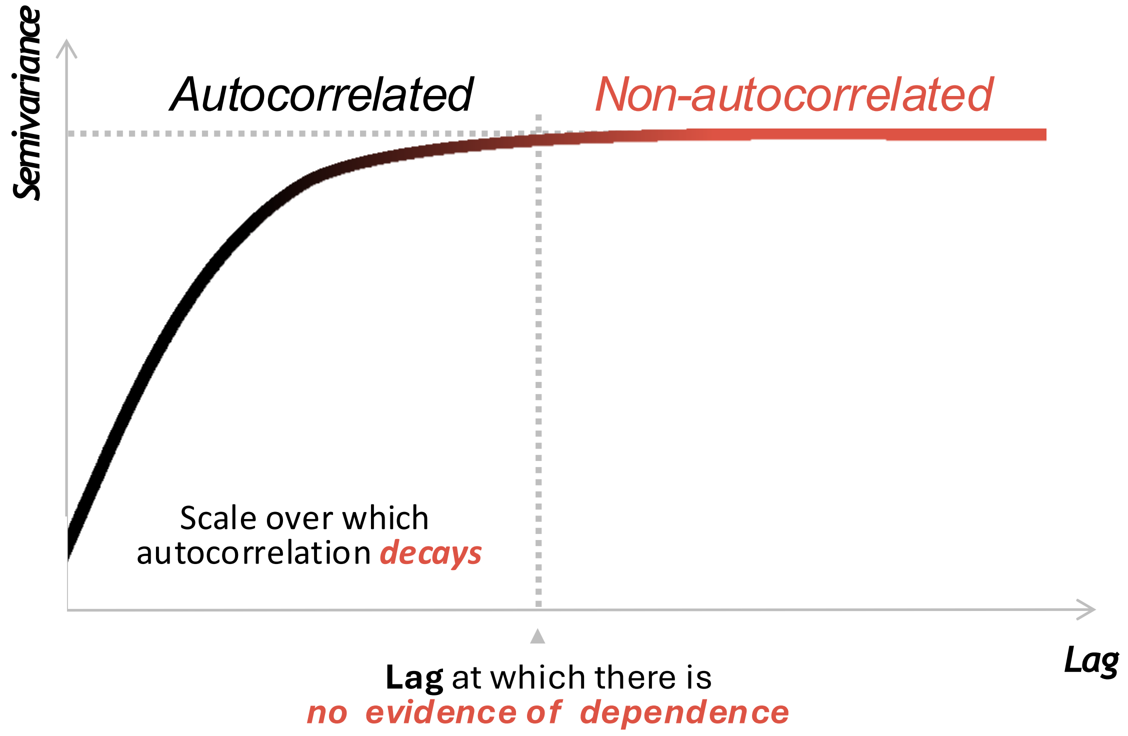

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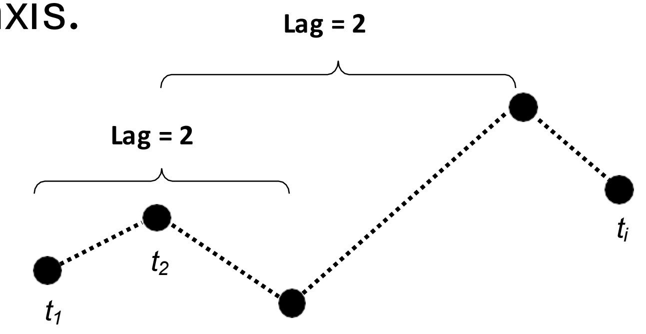

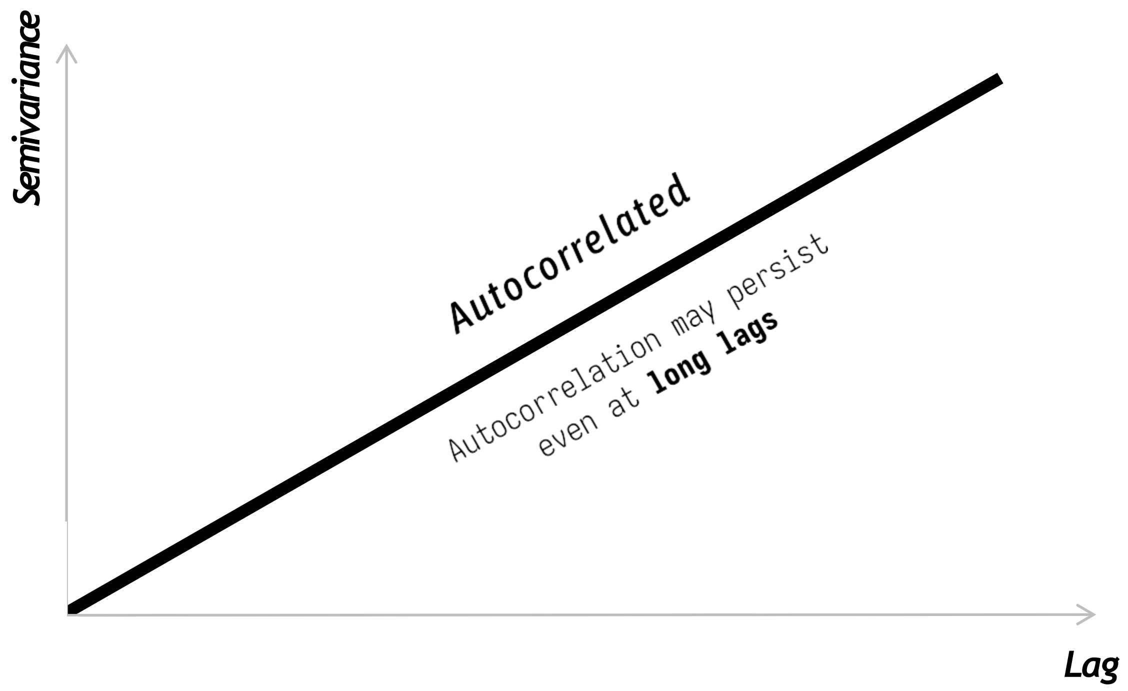

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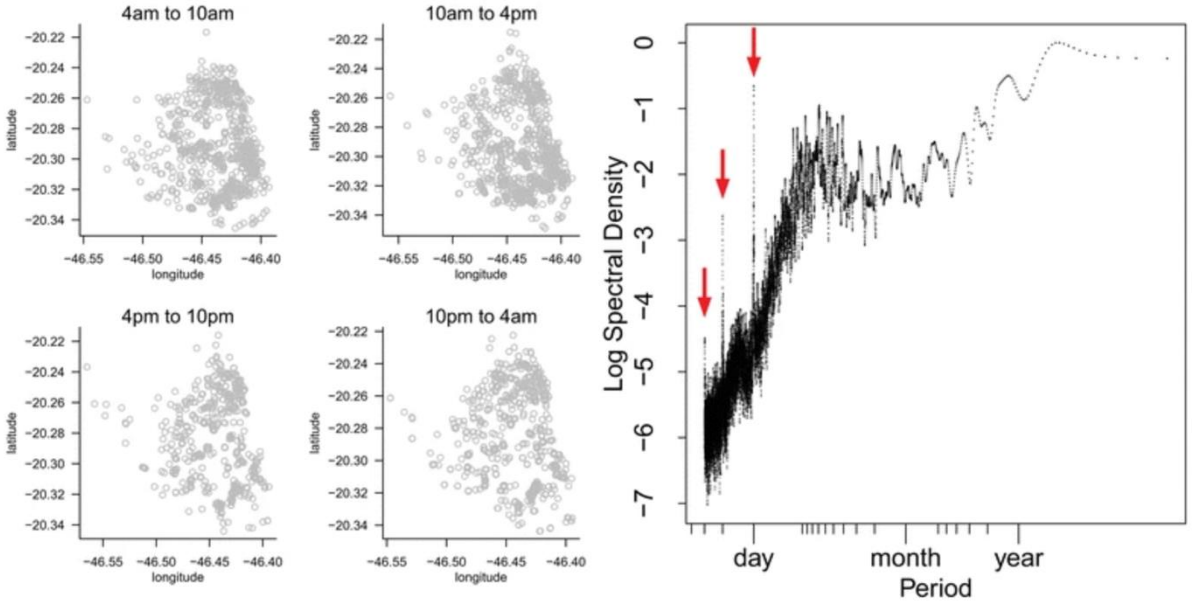
How can we measure and visualize autocorrelation?

- Autocorrelation in tracking data can also be visualized in terms of the spectral density of its variance

Maned Wolf
(Chrysocyon brachyurus)

- **Periodogram**:
  decomposition of variance to identify prominent frequencies,
  log spectral density as a function of the period (1/frequency)
  - Shows data across different time scales
  - Useful for patterns in movement periodicity



Péron *et al*. (2016)

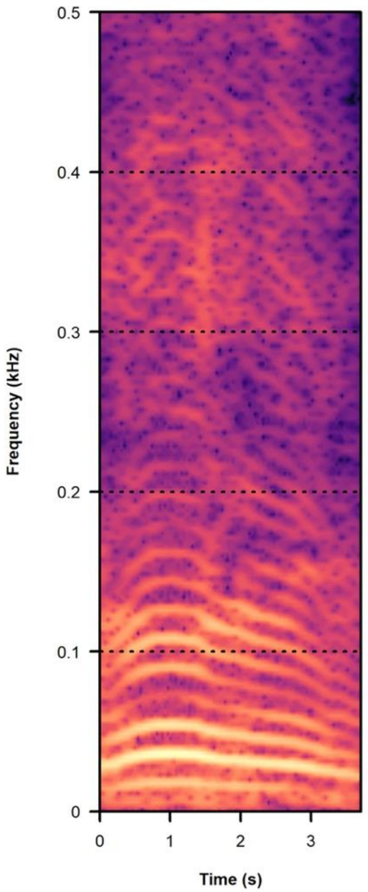
How can we measure and visualize autocorrelation?

- Autocorrelation in tracking data can also be visualized in terms of the spectral density of its variance

- **Spectrogram**:
  heat map of energy (intensity) of movement, frequency as a function of absolute time scale
  - Less common in movement ecology
  - But useful for visualizing movement patterns

African Savanna Elephant
(Loxodonta africana)

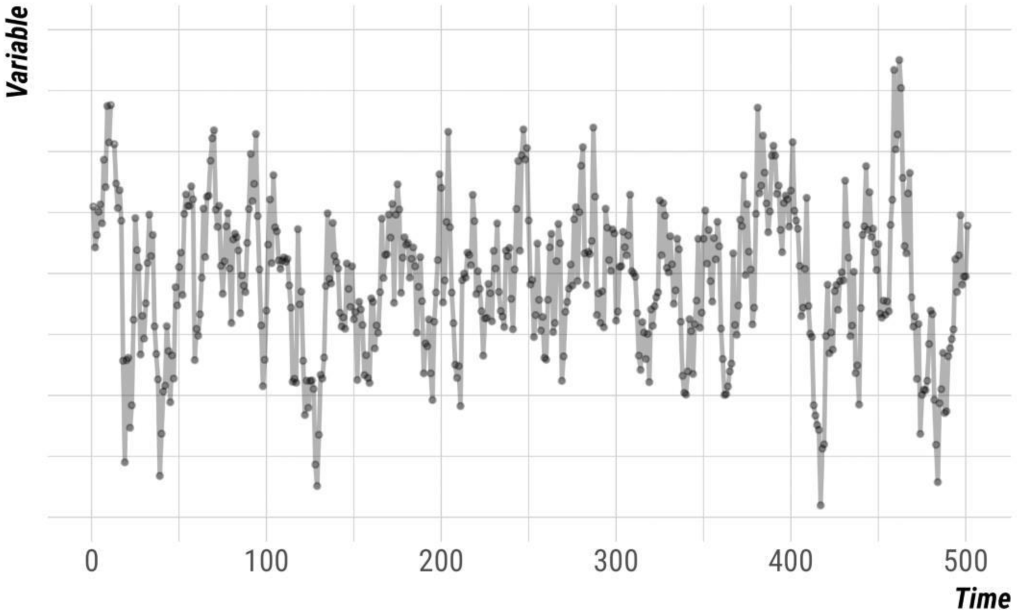
Pardo *et al.* (2024)
DOI: 10.1038/s41559-024-02420-w

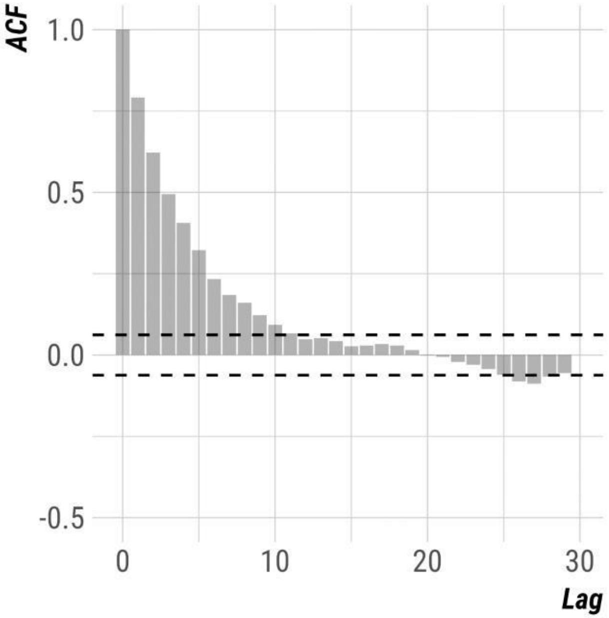


Autocorrelation

- The *sampling frequency* (or *duration*) of data collection can influence our ability to detect autocorrelation

Φ = 0.8





Detecting autocorrelation

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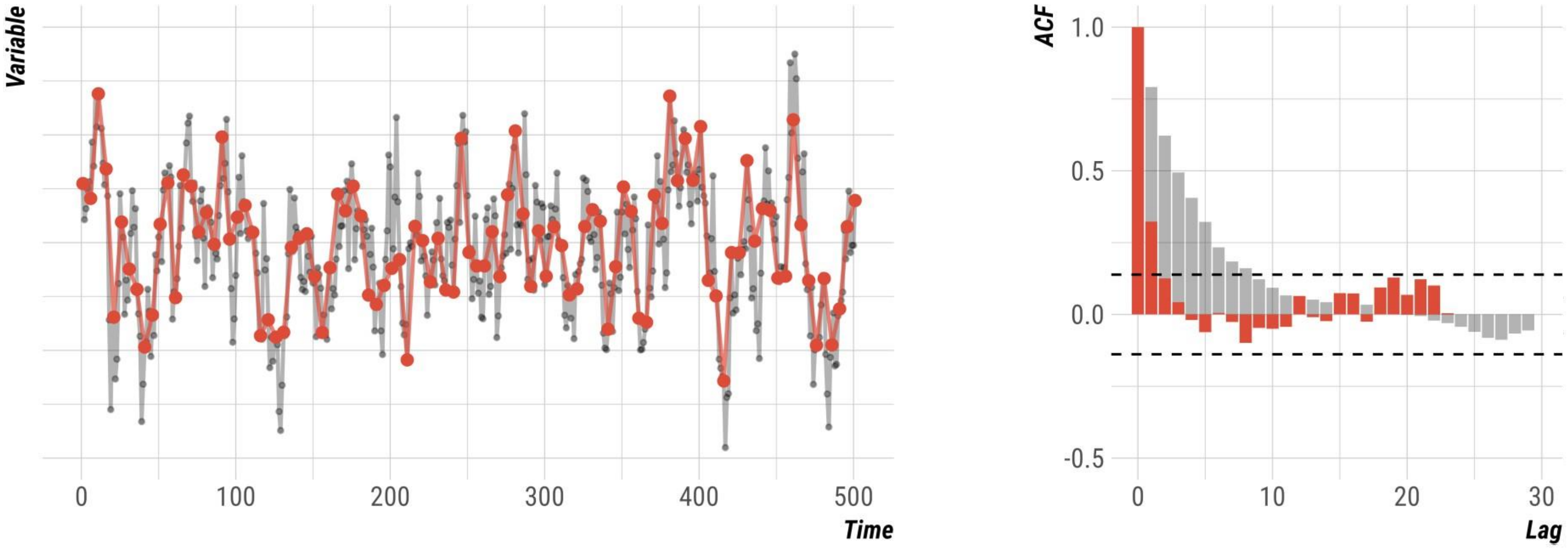

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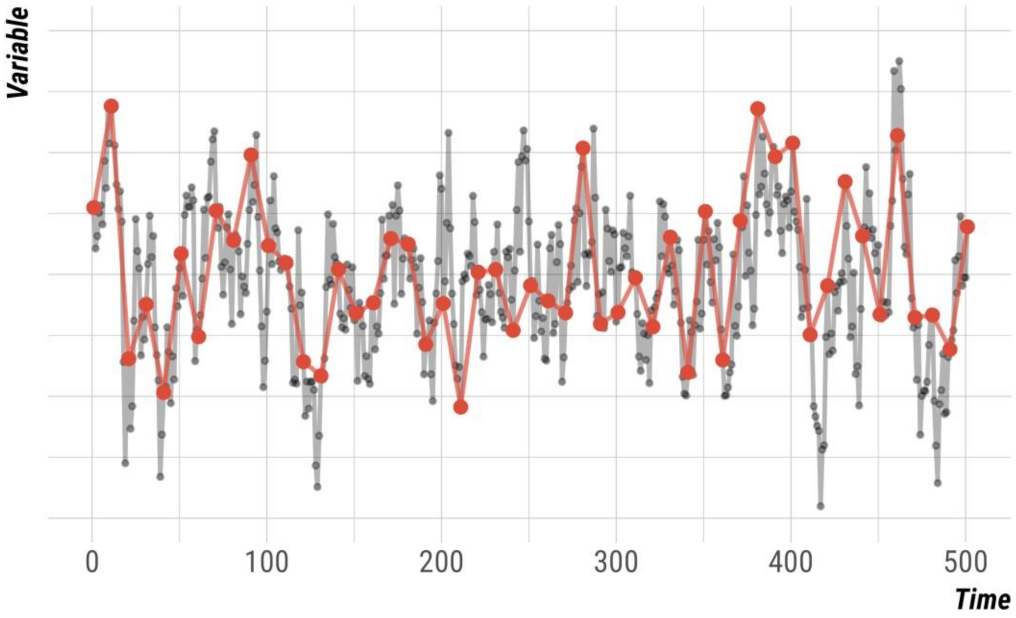

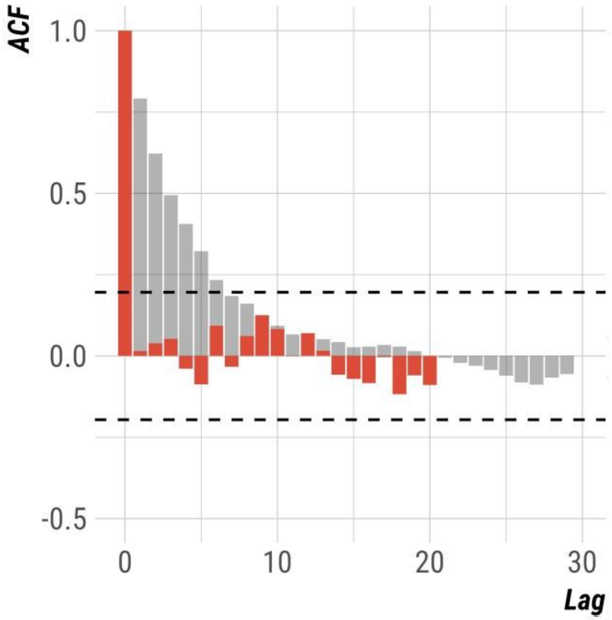

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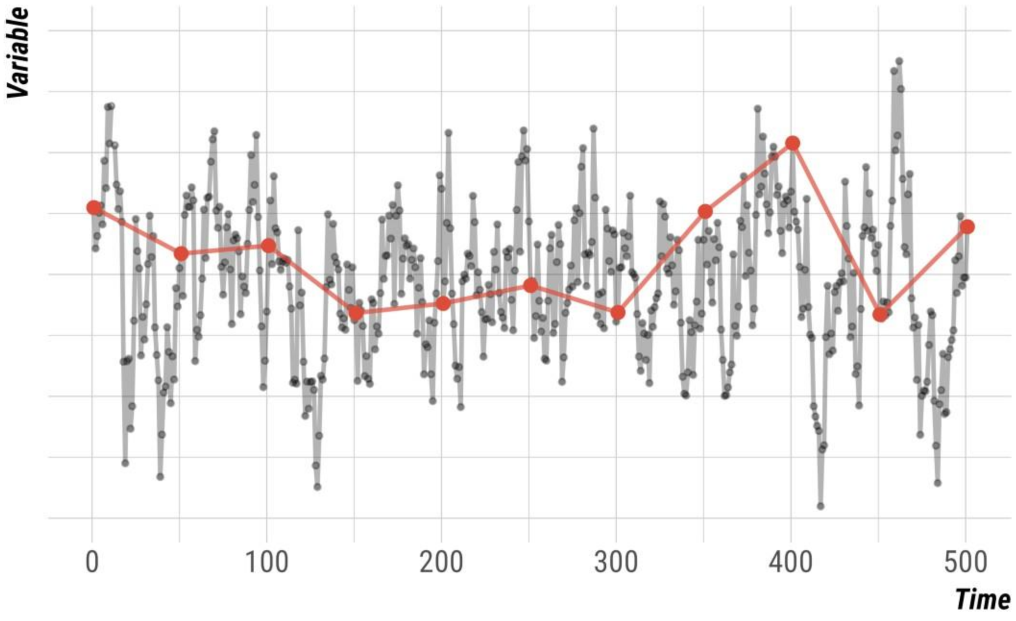

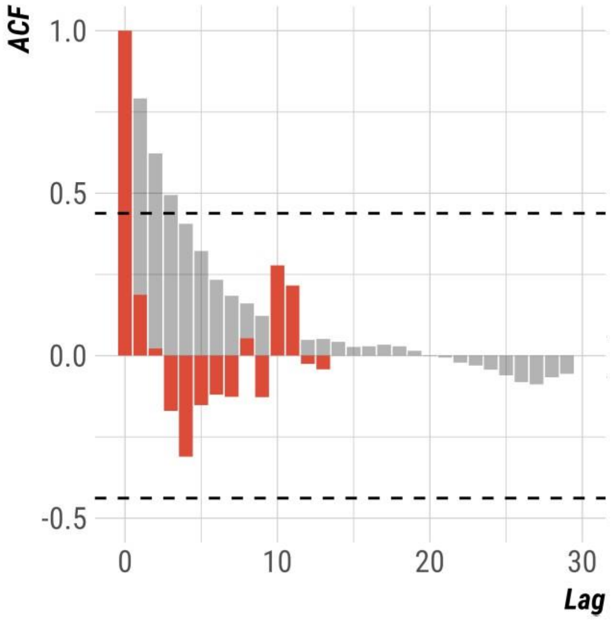

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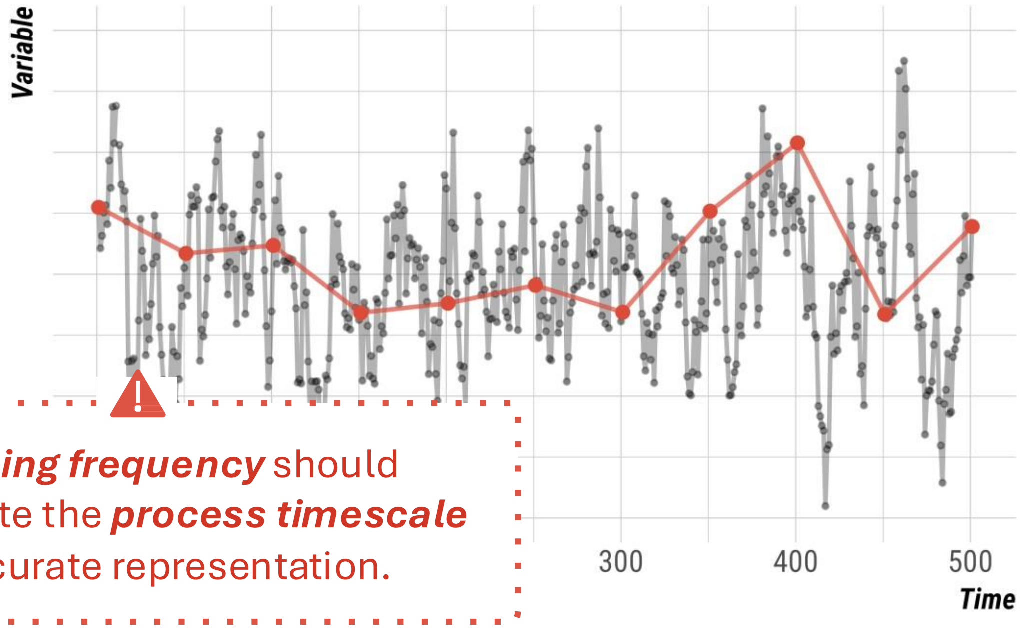
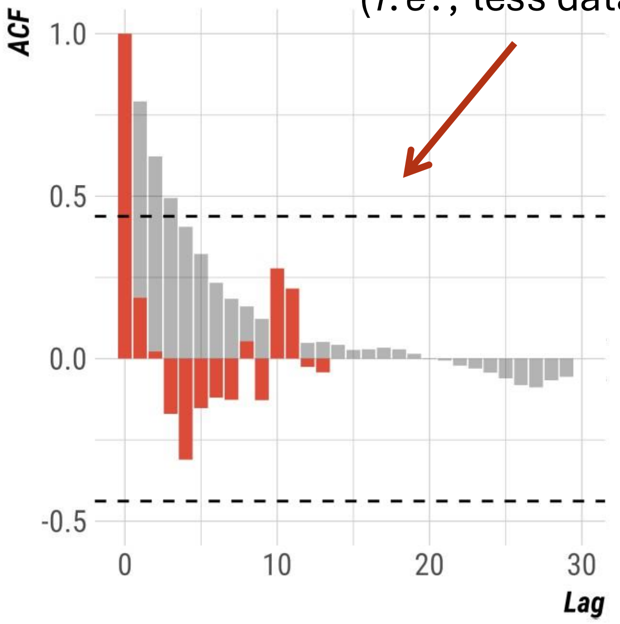


Sampling frequency should encapsulate the ***process timescale*** for accurate representation.

Confidence intervals increase because sample size decreases ($i.e.$, less data points).



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Animal tracking data are a time-series

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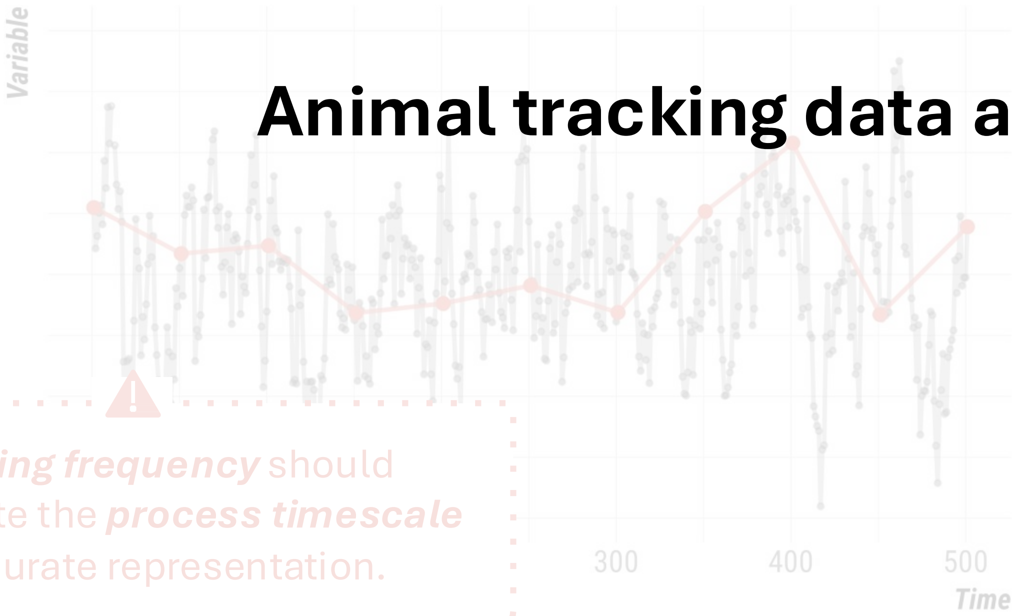

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The impact of autocorrelation

- *Sample size*, *n* is the denominator when calculating independent and identically distributed (IID) SEs and CIs

Standard error (SE):

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}}$$

Confidence intervals (CI):

$$95\%\ \mathrm{CI} = \hat{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

All else equal, as *sample size increases*, both *SE* and *CI width decrease*

- With autocorrelated data, *each new datapoint* is related to a previously collected datapoint and *does not bring a full independent datapoint* worth of information
- e.g., 90% autocorrelation ≈ 10% new info

The impact of autocorrelation

- When data are autocorrelated, SEs and CEs are underestimated
  - Effect is usually strongest on SEs and CIs, but can also impact the mean

Why does this matter?

- Ignoring autocorrelation can lead to overly optimistic results
  - Underestimating uncertainty —researchers might conclude that their estimates are more precise than they truly are
  - Misleading statistical inferences —incorrect conclusions about significance

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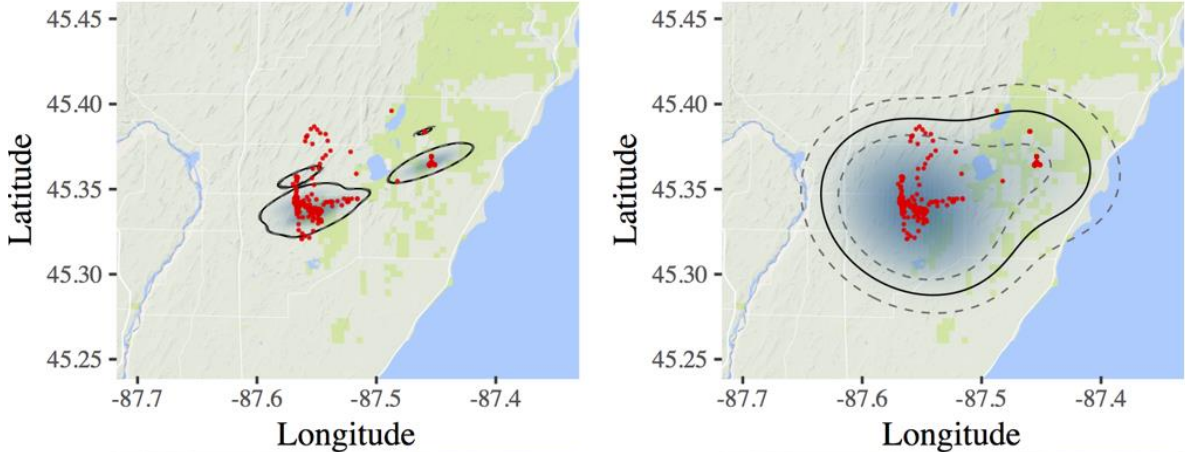
Motivating example:
Neglecting autocorrelation in home-range estimation



- GPS-tracked black bear

Black Bear
(Ursus americanus)

(Noonan, Tucker, Fleming, et al., Ecological Monographs. 2019)

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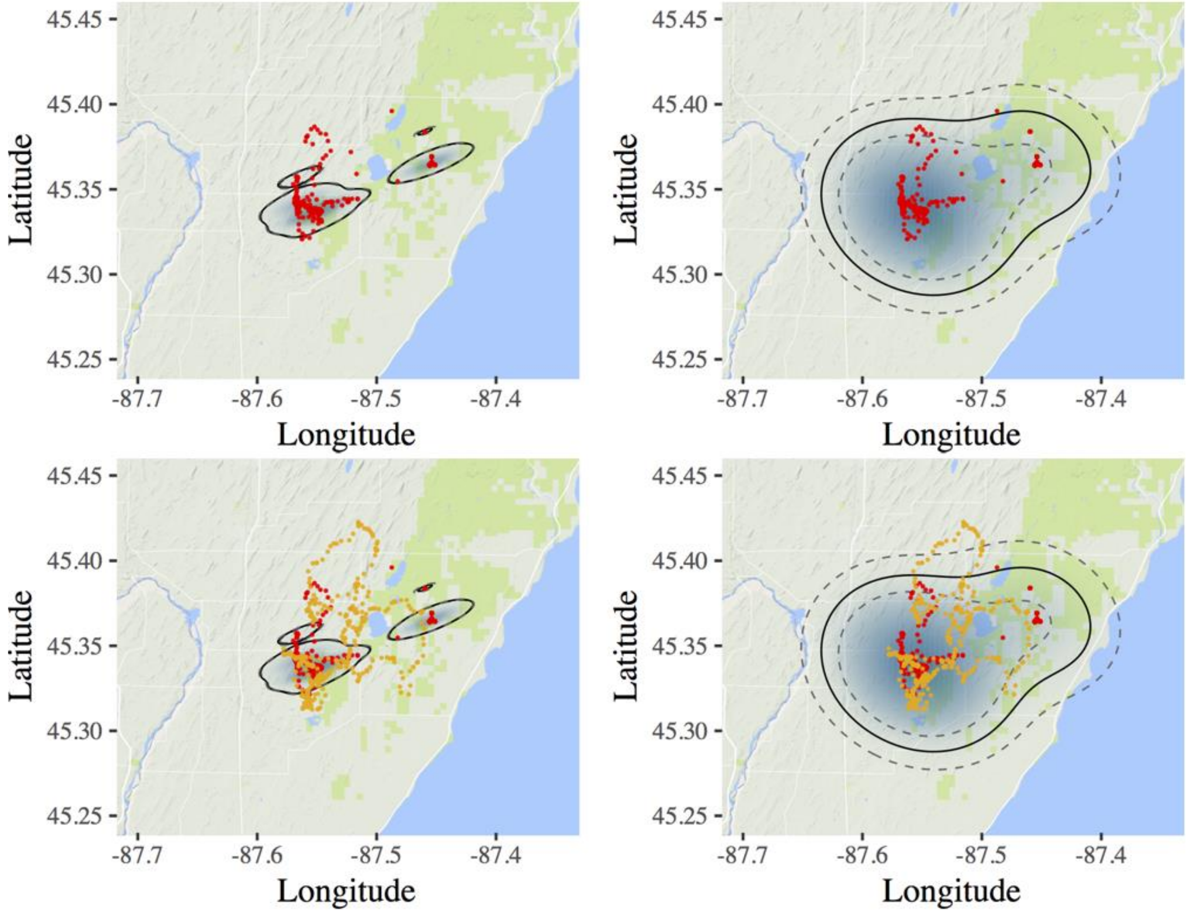

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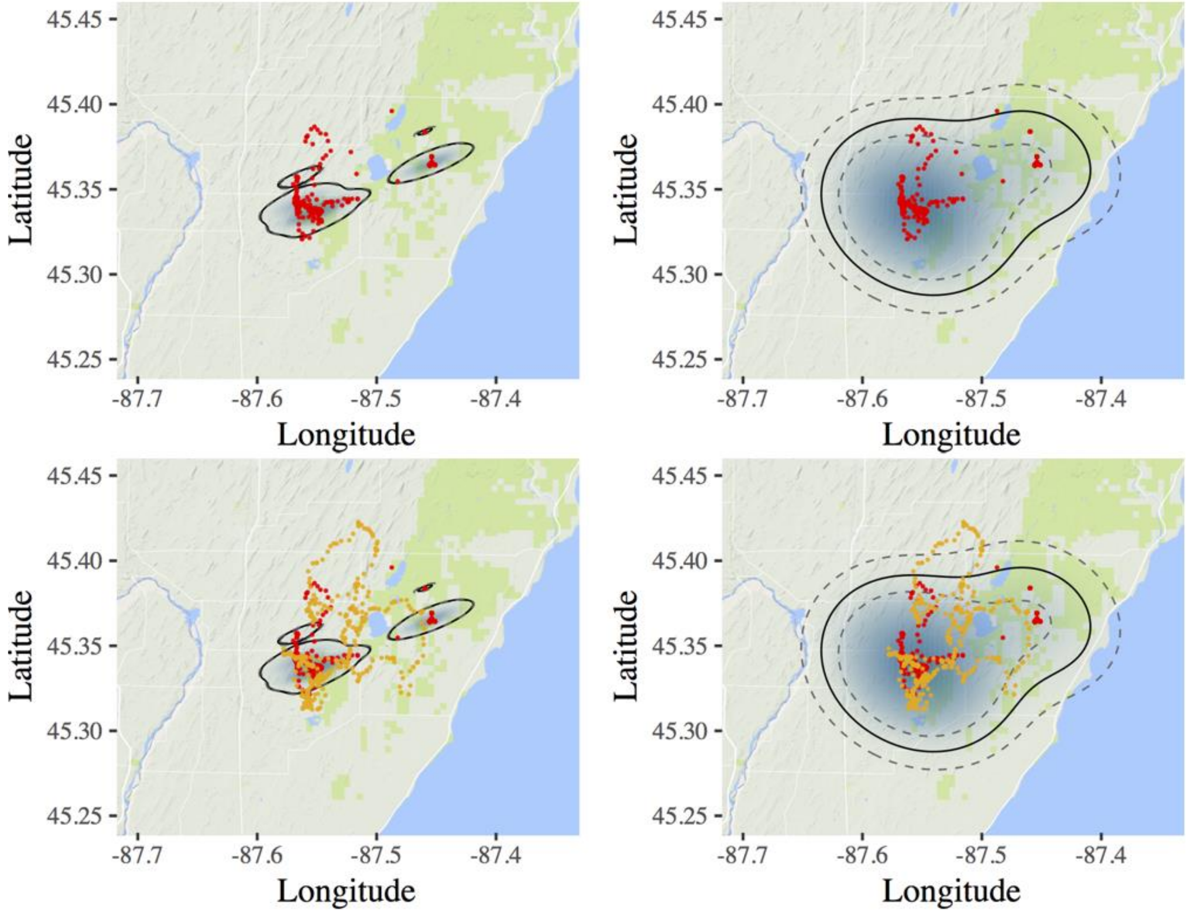

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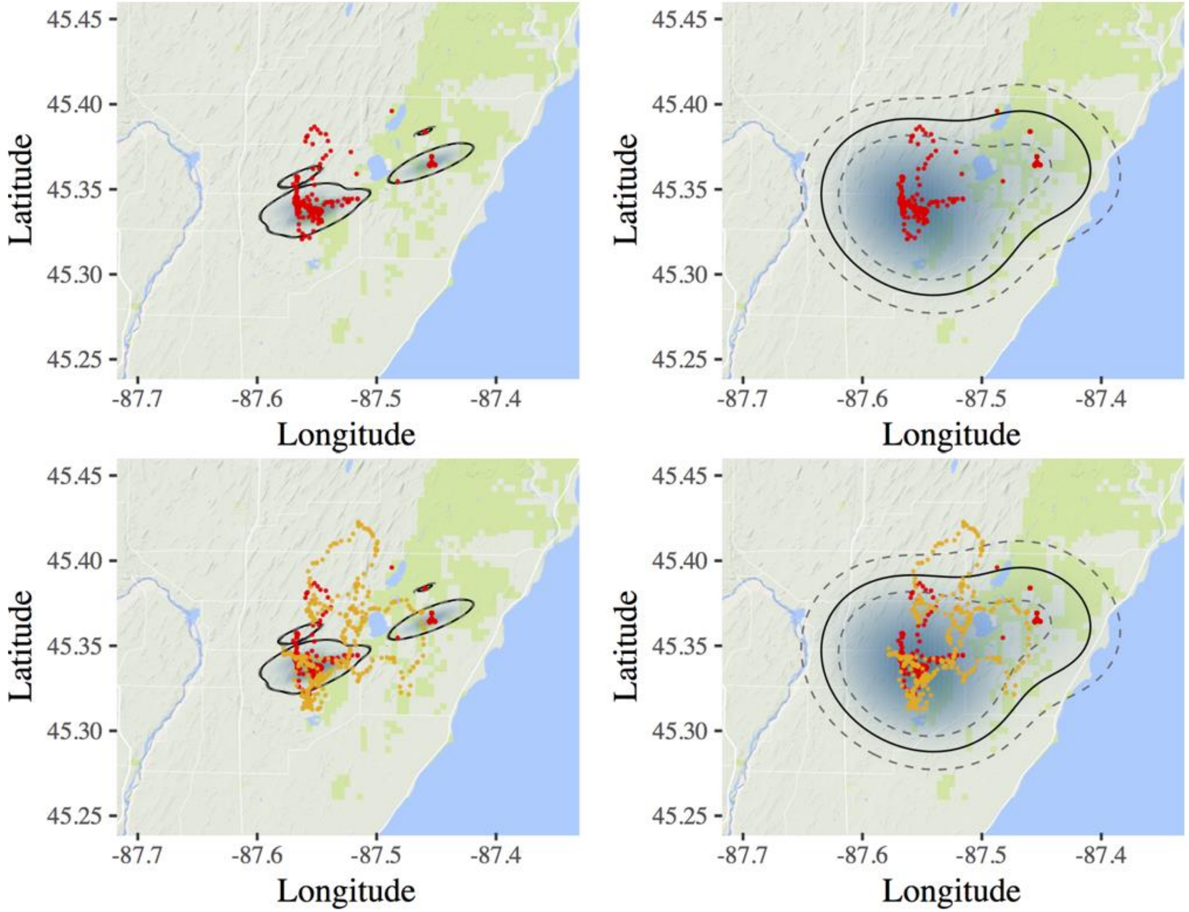

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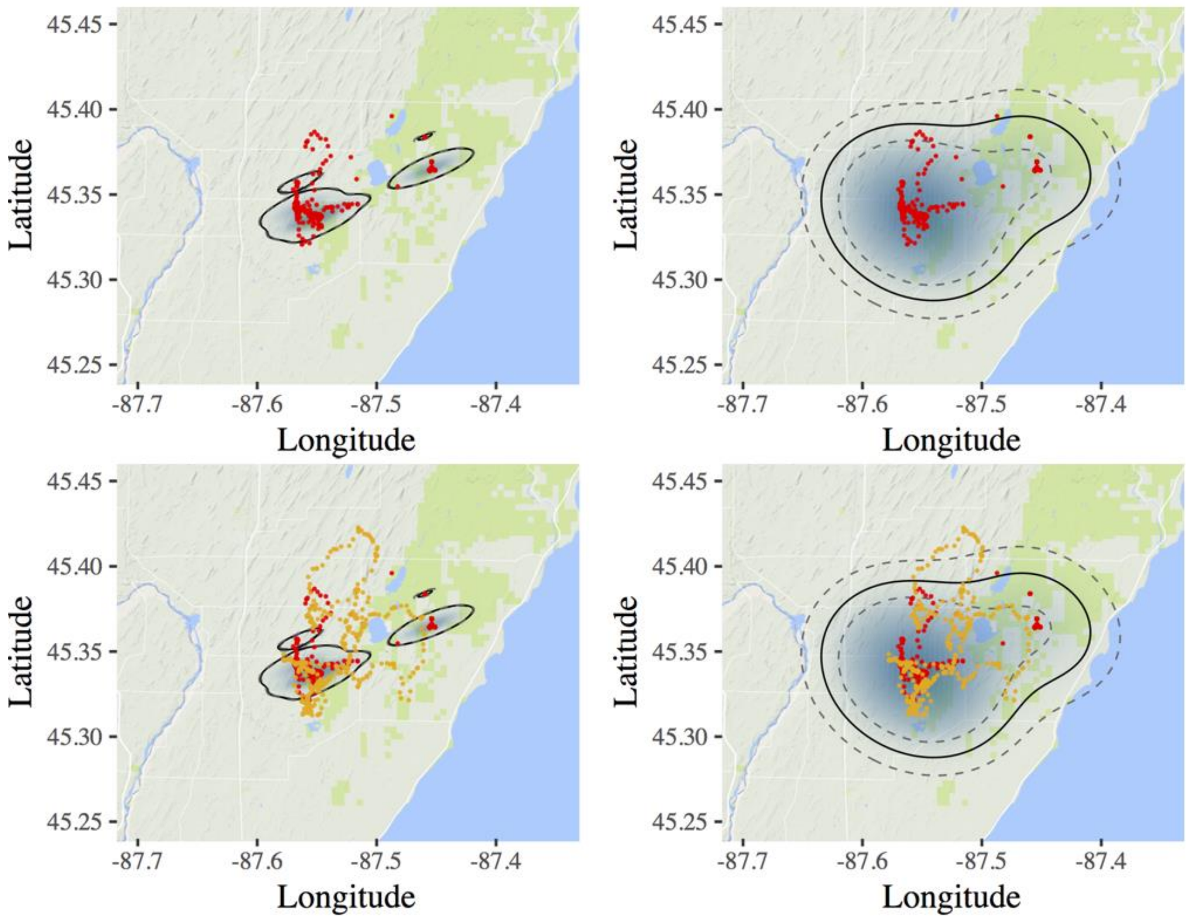

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- Q2:
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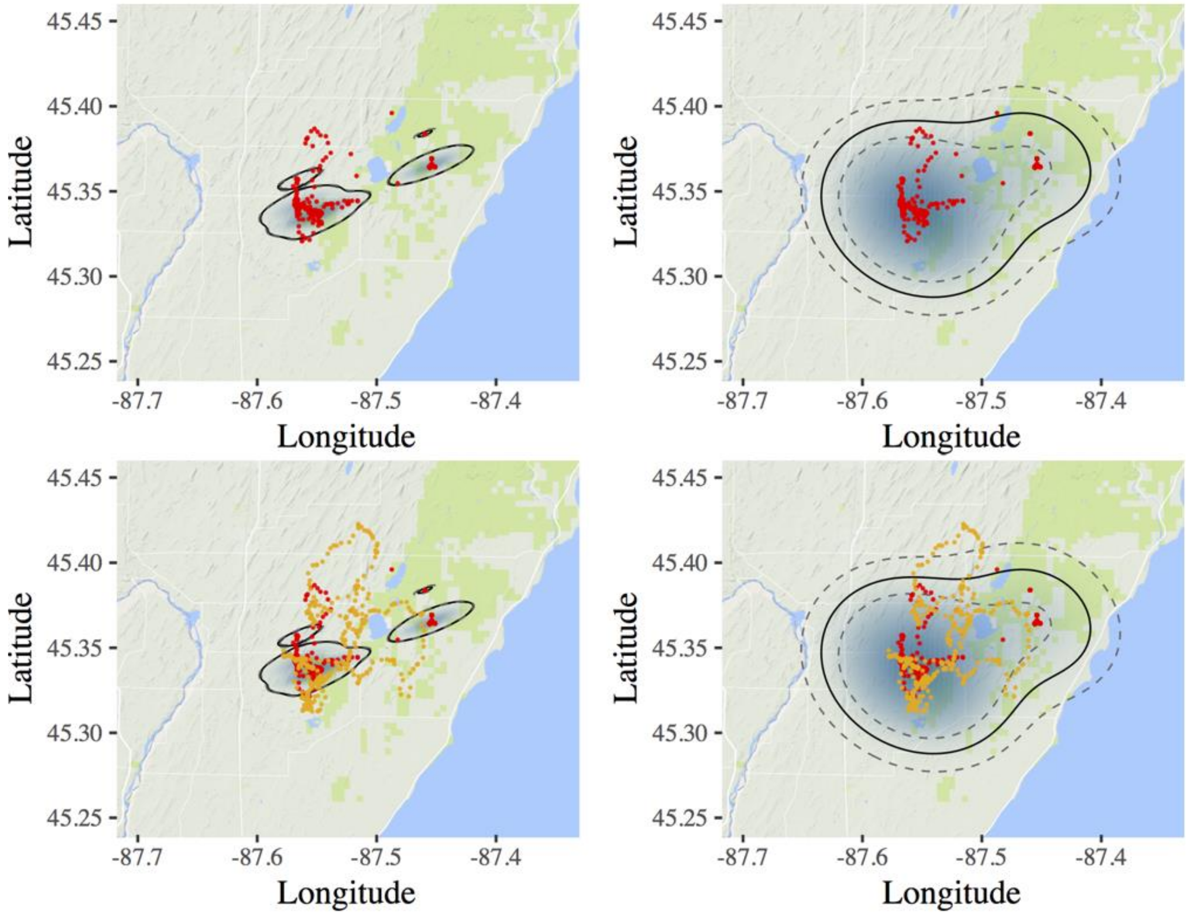

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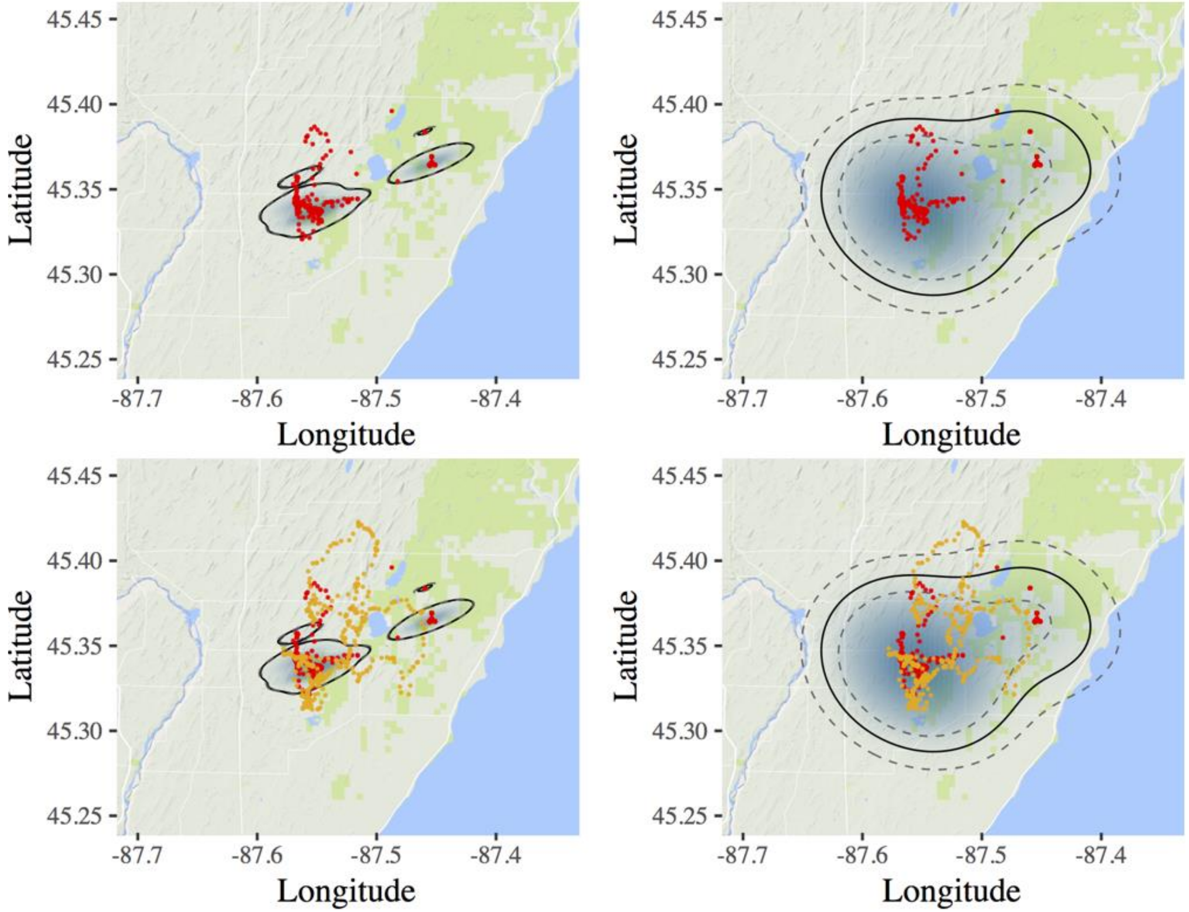

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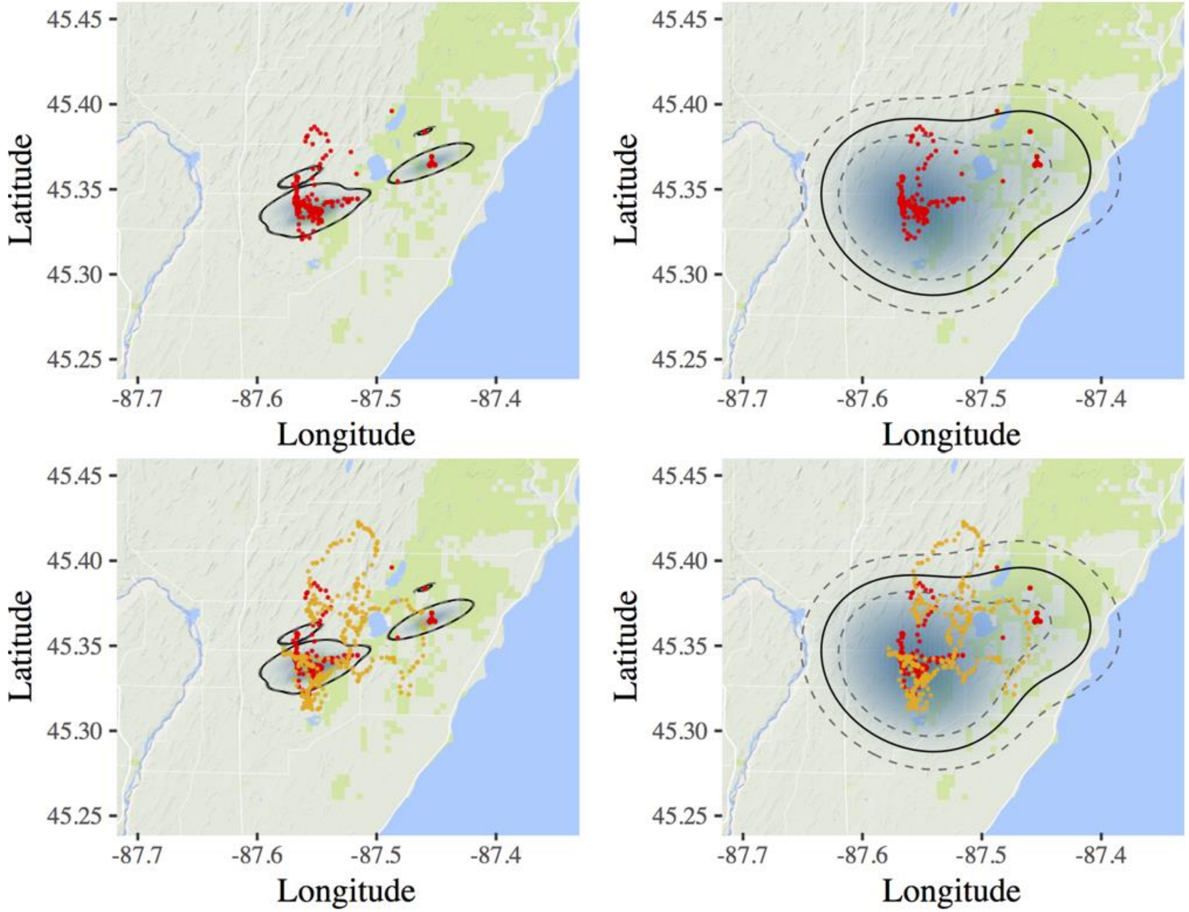

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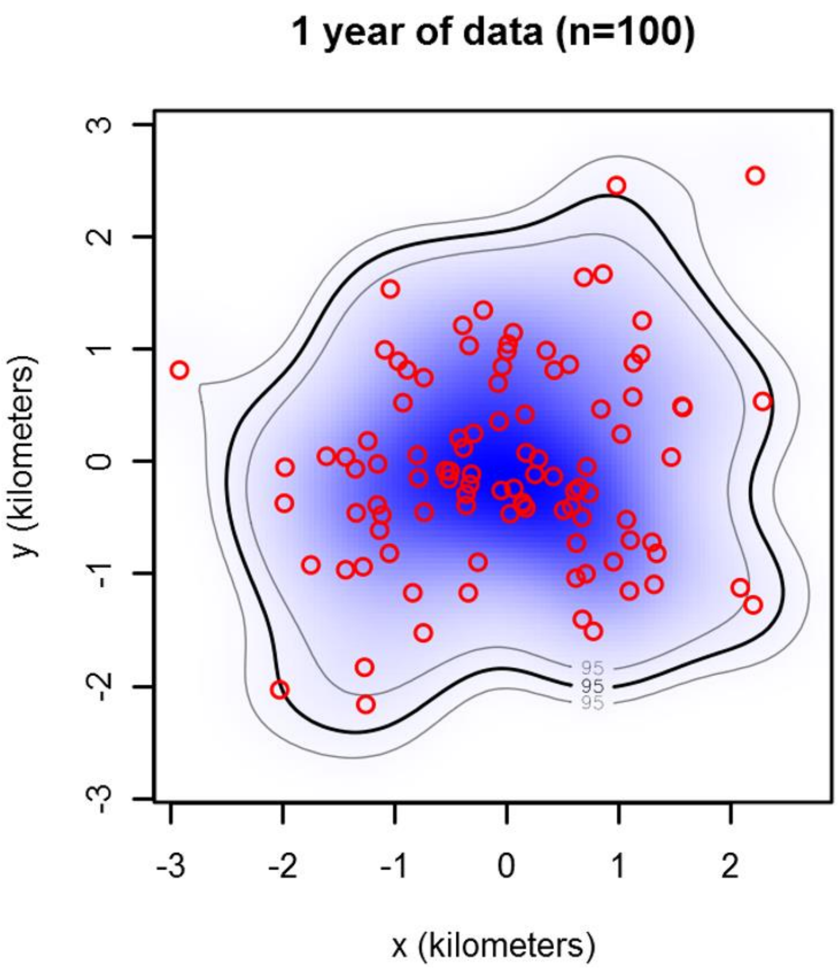
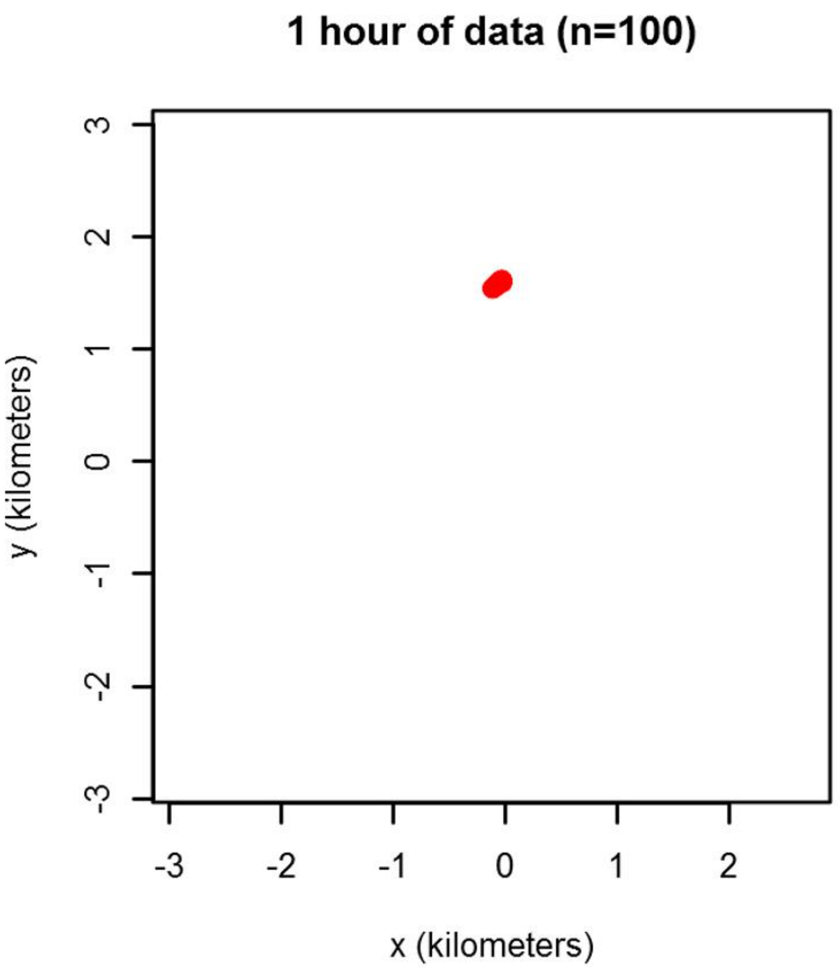
- Q2: Does this happen in practice?
- Q3: How did we fix this?

Q1: Why does this happen?





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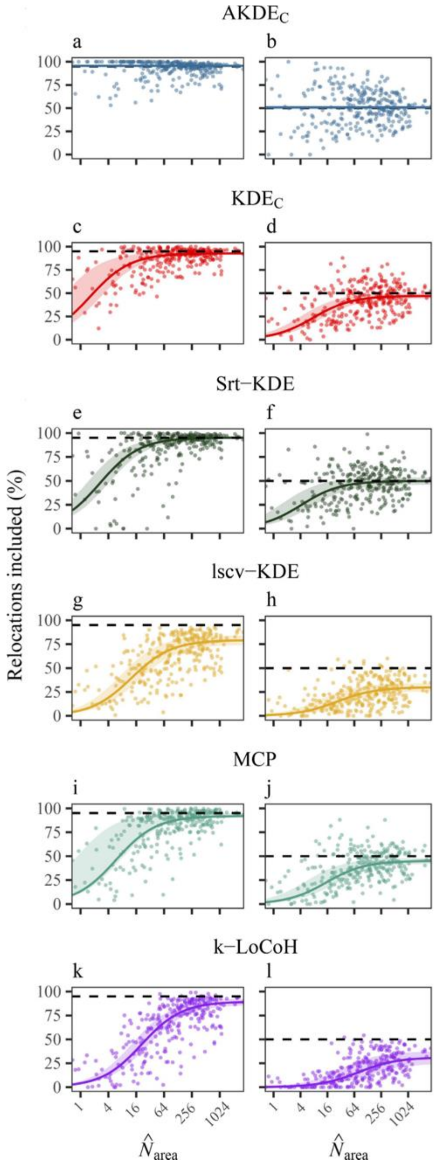
- **Conventional estimates are 2–20 times too small**, on average

Q2: Does this happen in practice?



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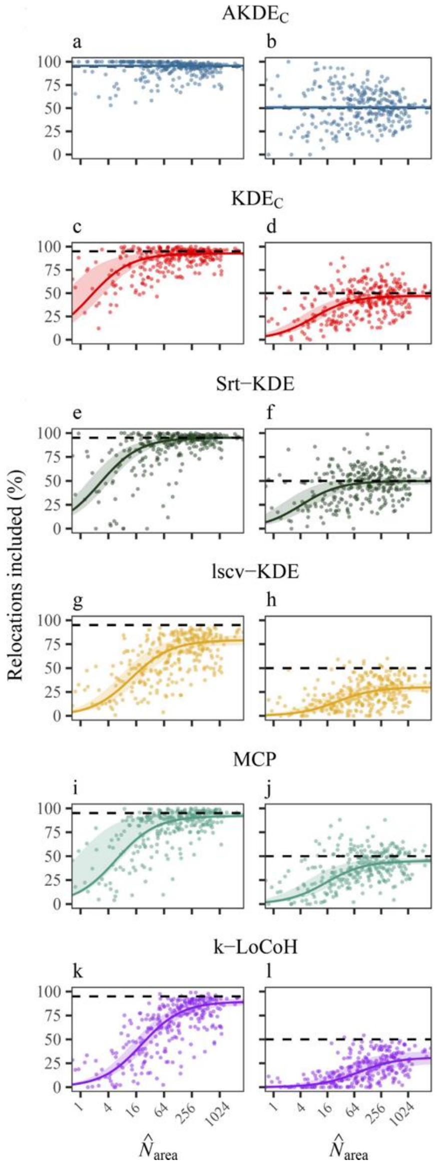

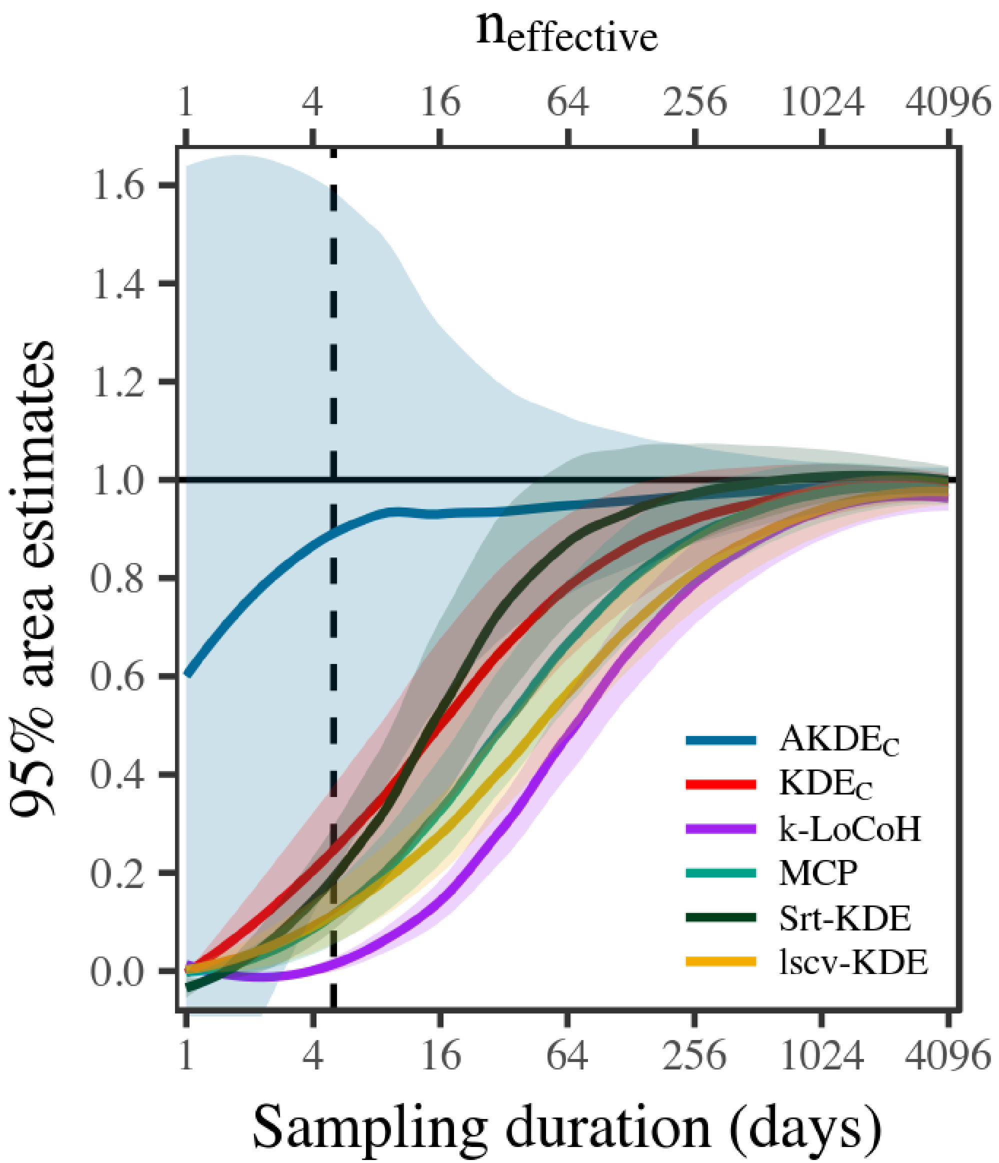

Q2: Does this happen in practice?

- Kernel Density Estimation (KDE) is a statistically optimal non-parametric method for estimating distributions

  — for independently sampled data
- Generalized to autocorrelated data
  - Fleming et al., Ecology (2015); Fleming & Calabrese, Methods in Ecology and Evolution (2017); Fleming et al., Ecological Applications (2018)
- Largest comparative analysis to date: 369 individuals, 27 species
  - Noonan, Tucker, Fleming, et al., Ecological Monographs (2019)
- Conventional estimates are 2–20 times too small, on average
- **Bias is worse for larger species**
  - Noonan, Fleming, et al., Conservation Biology (2020)

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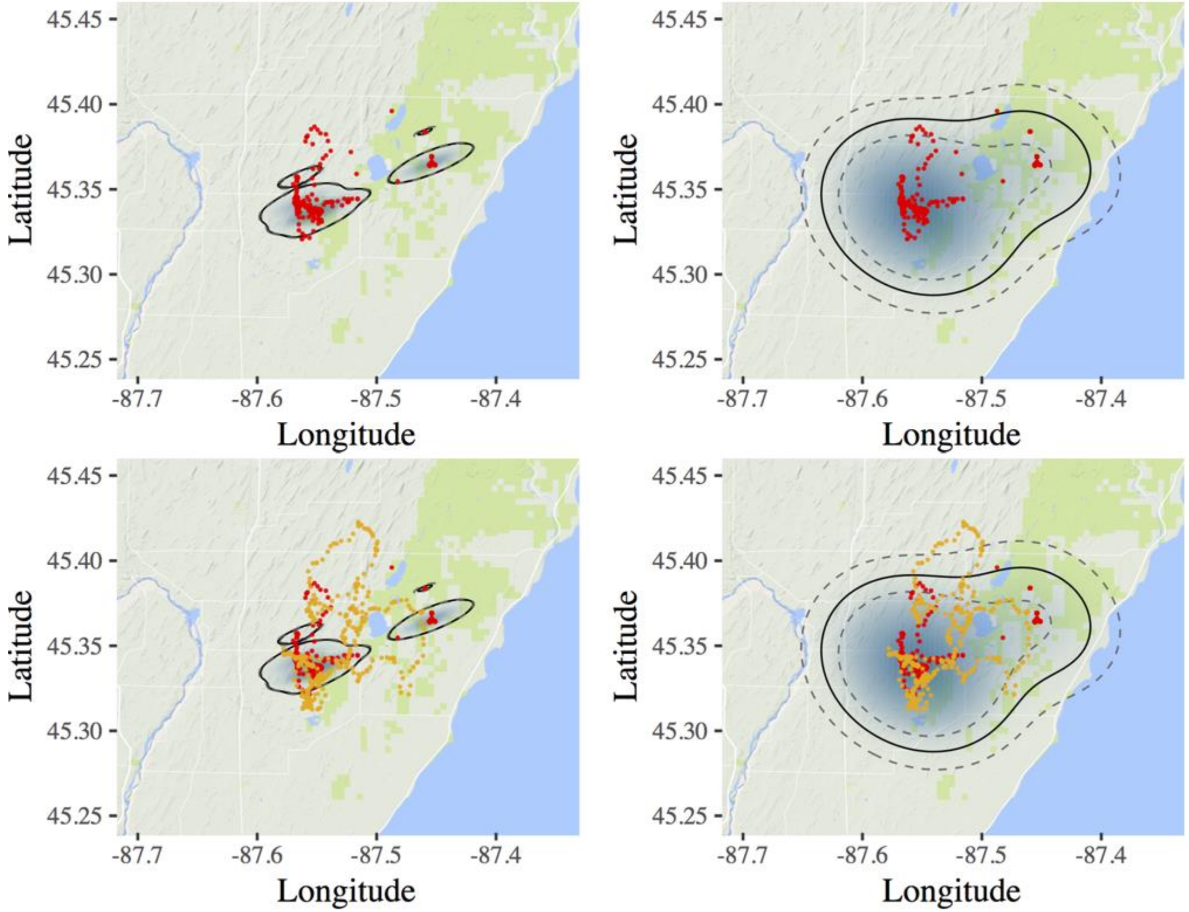
The underestimation of animal space use— The solution:



(Noonan, Tucker, Fleming, et al., Ecological Monographs. 2019)

- GPS-tracked black bear
- Ignoring non-independence leads to overconfidence
- Space use is underestimated
- Q1: Why does this happen?
- Q2: Does this happen in practice?
- Q3: How did we fix this?

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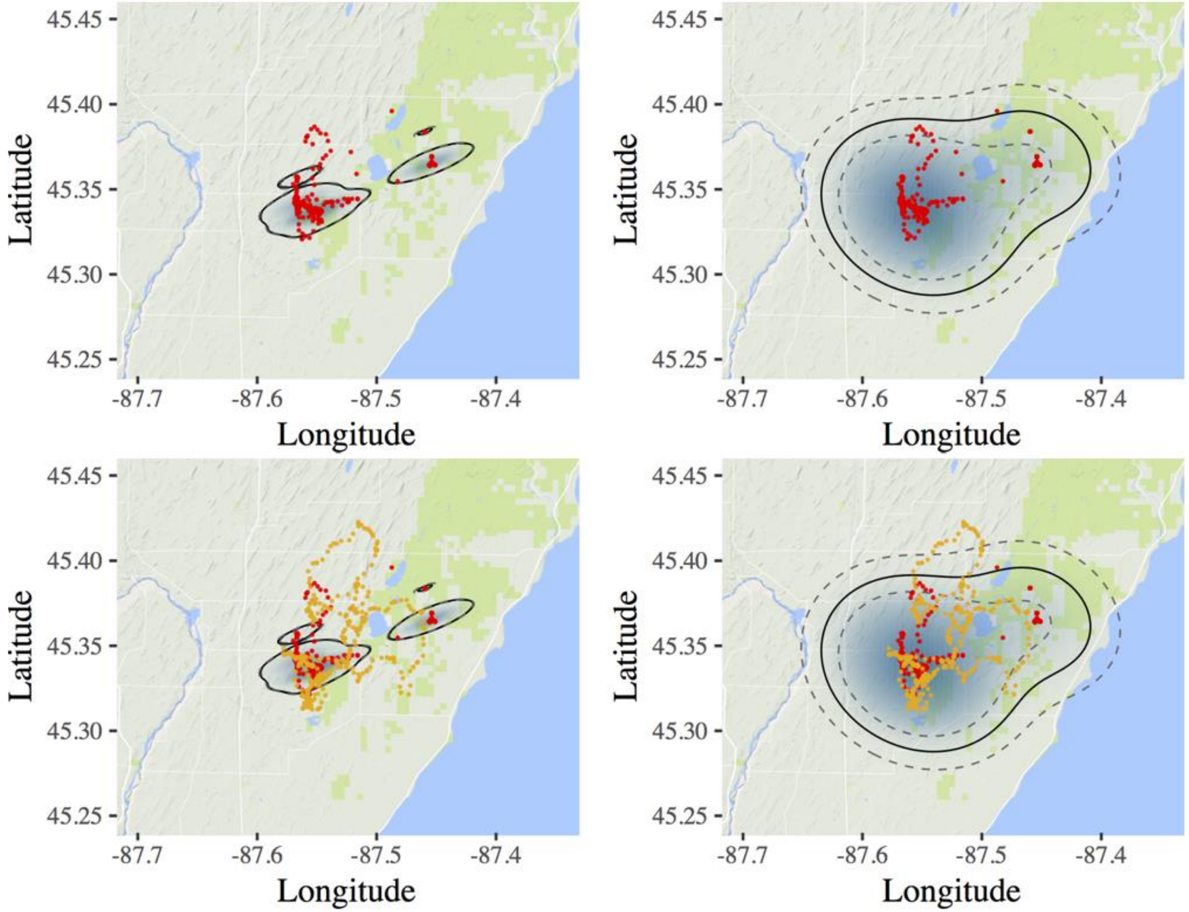

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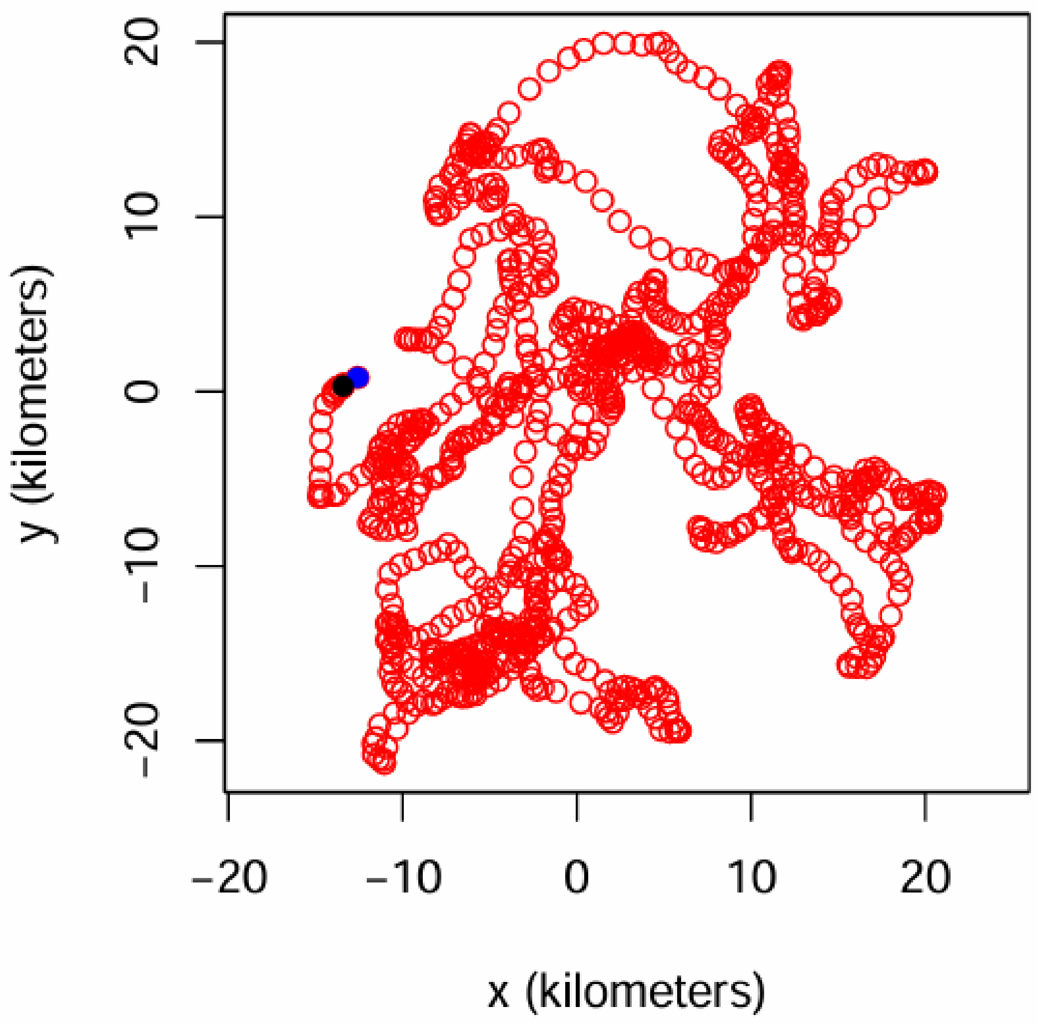
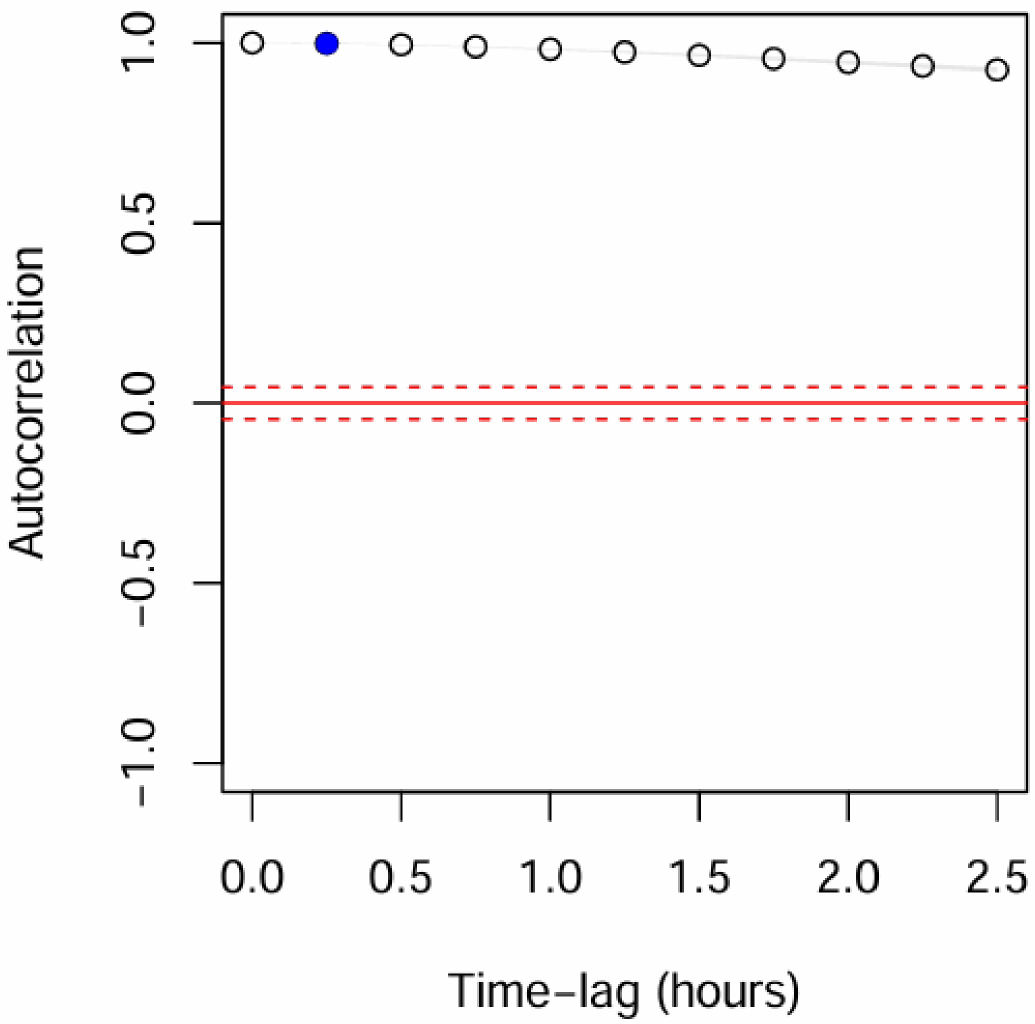
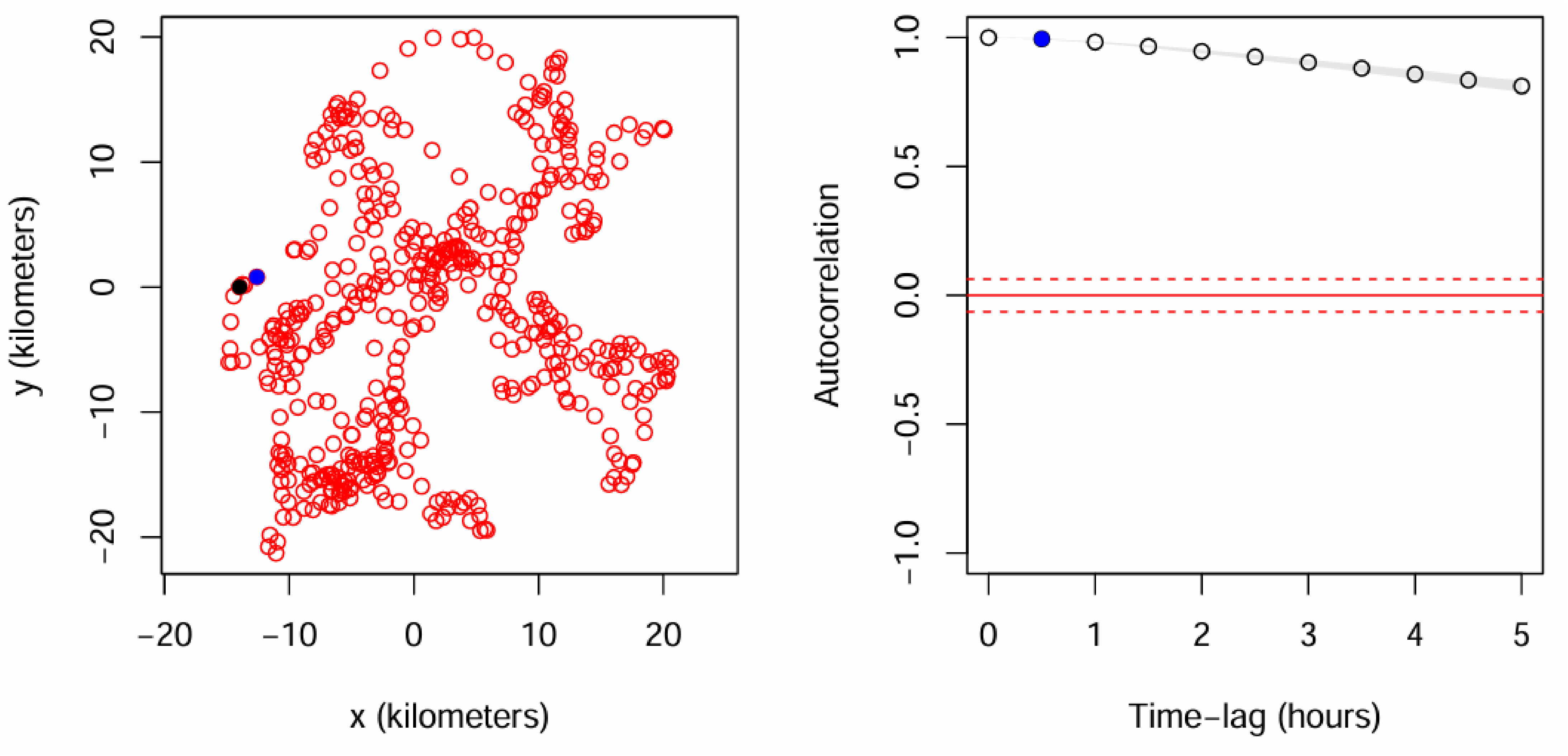
- A3: Continuous-time stochastic process models of the ***autocorrelation***

autocorrelation is





If autocorrelation is the culprit…



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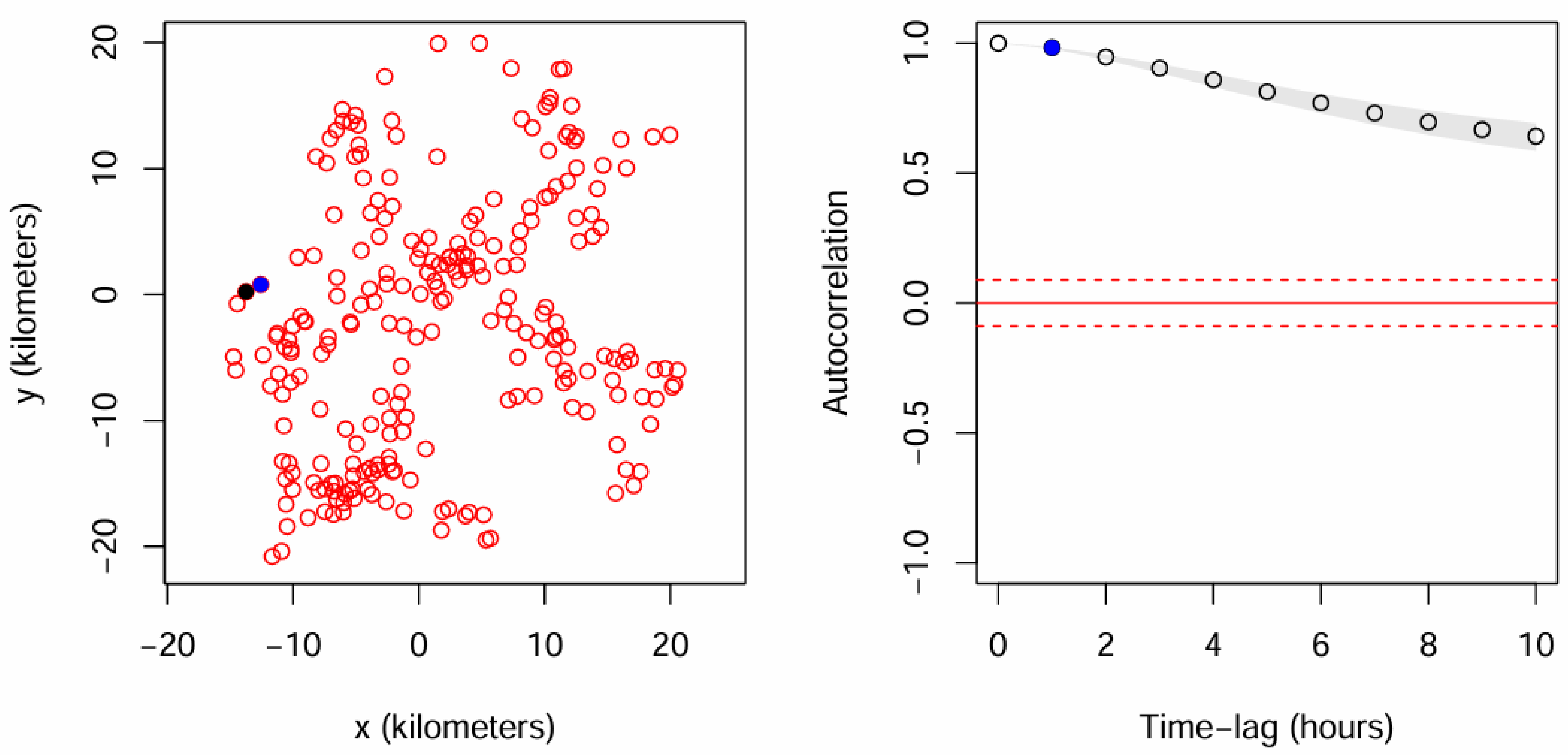

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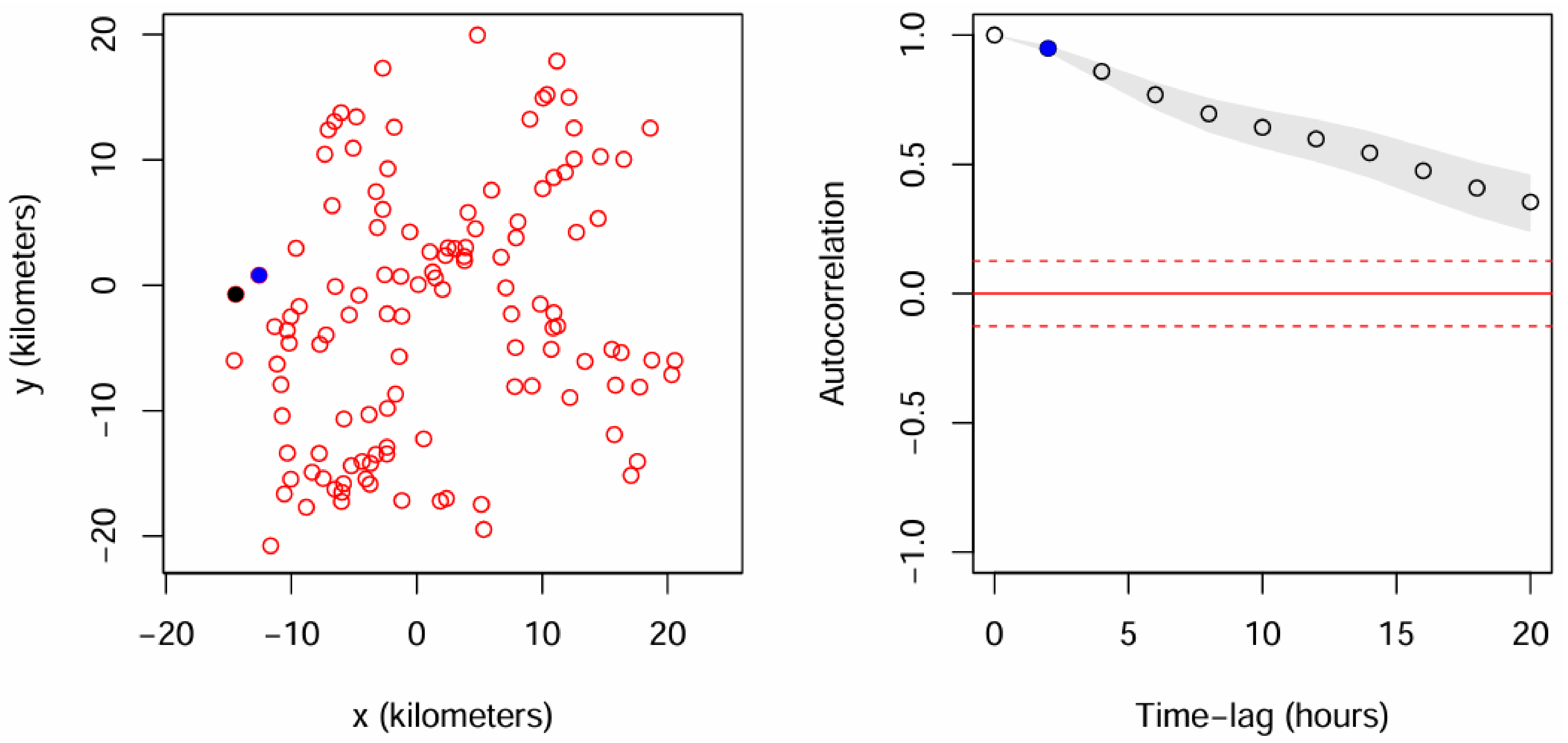

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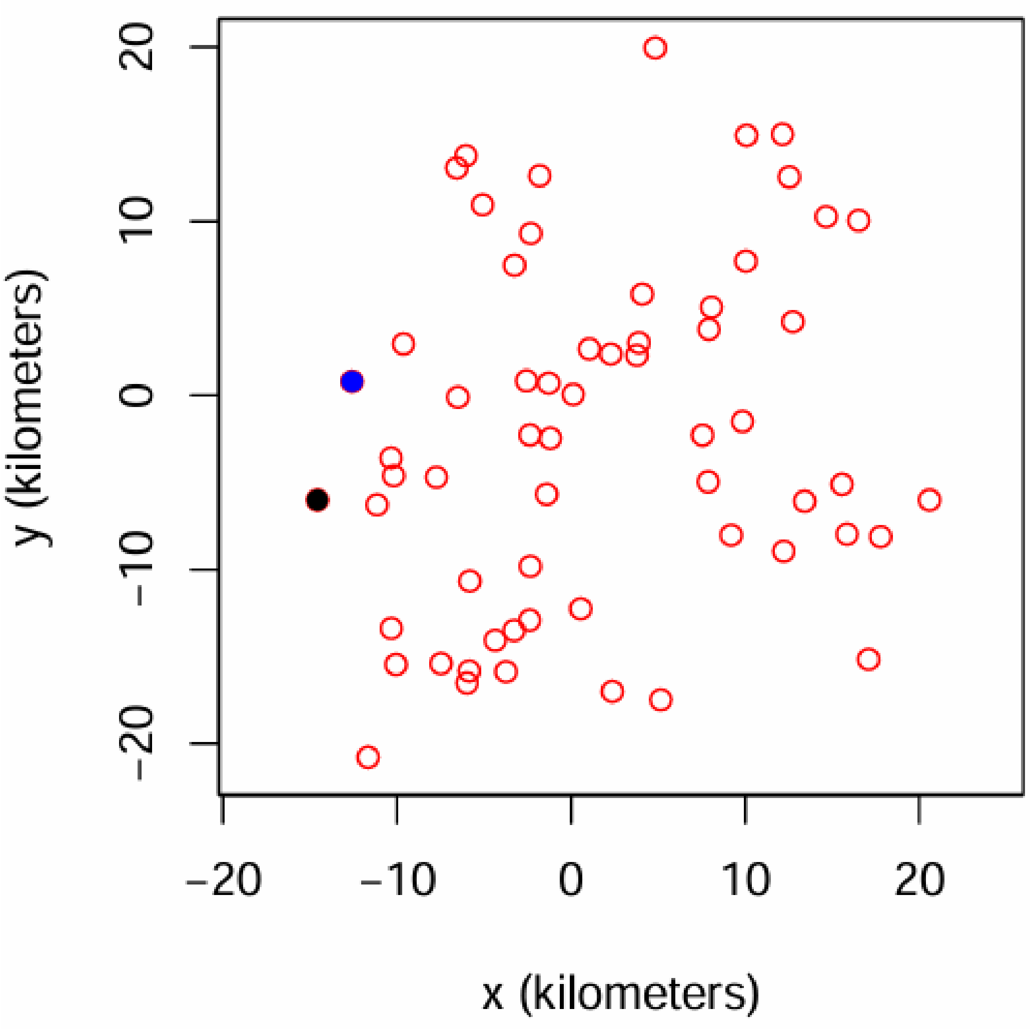

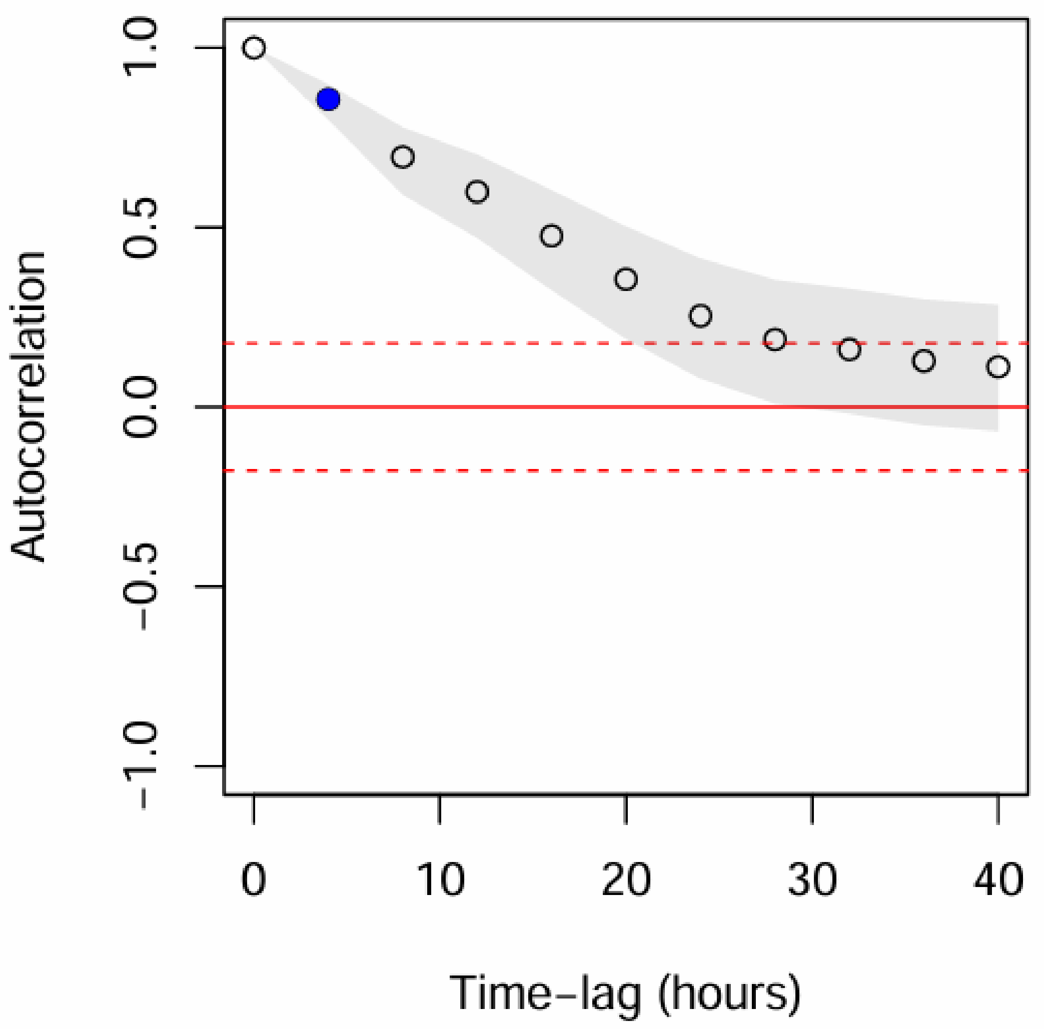

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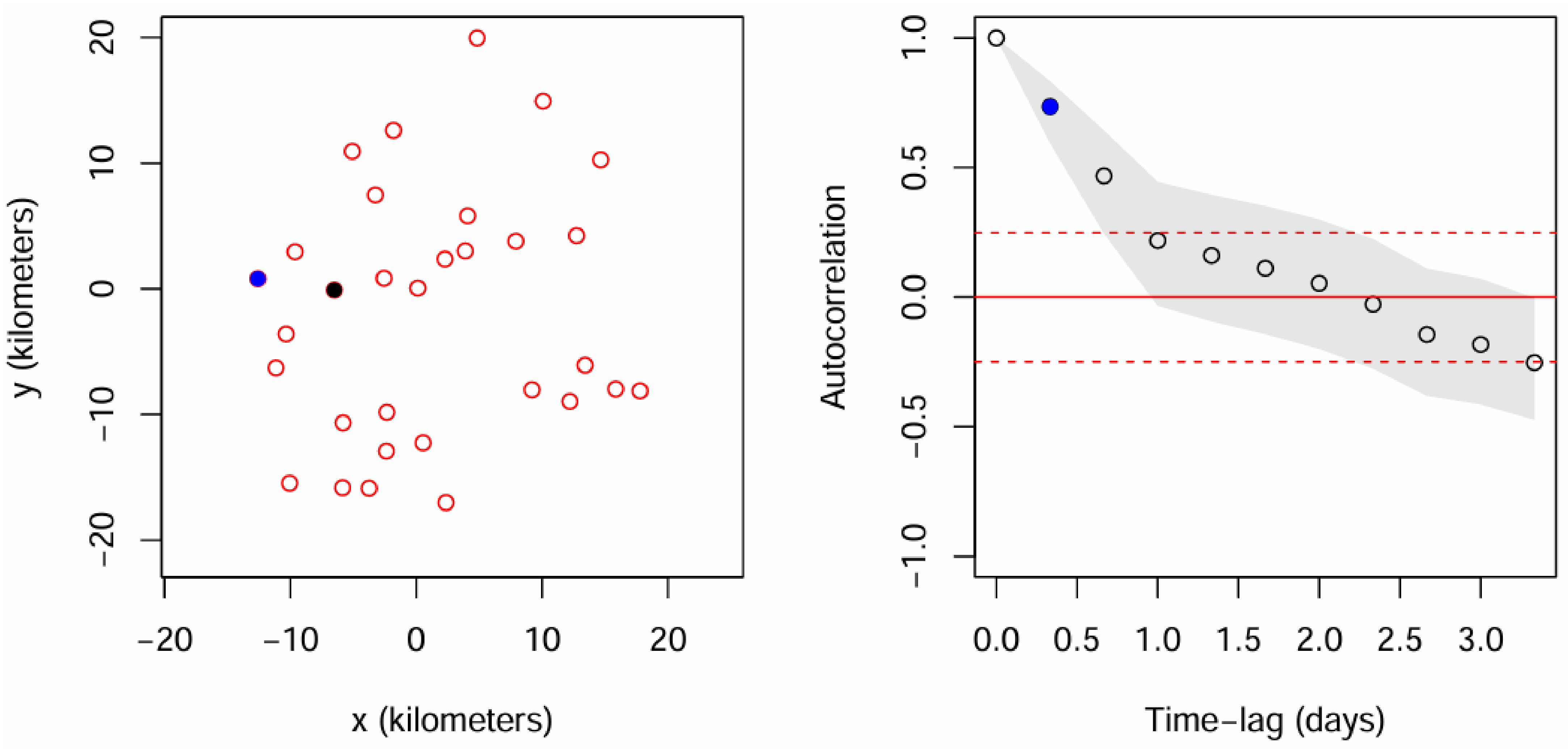

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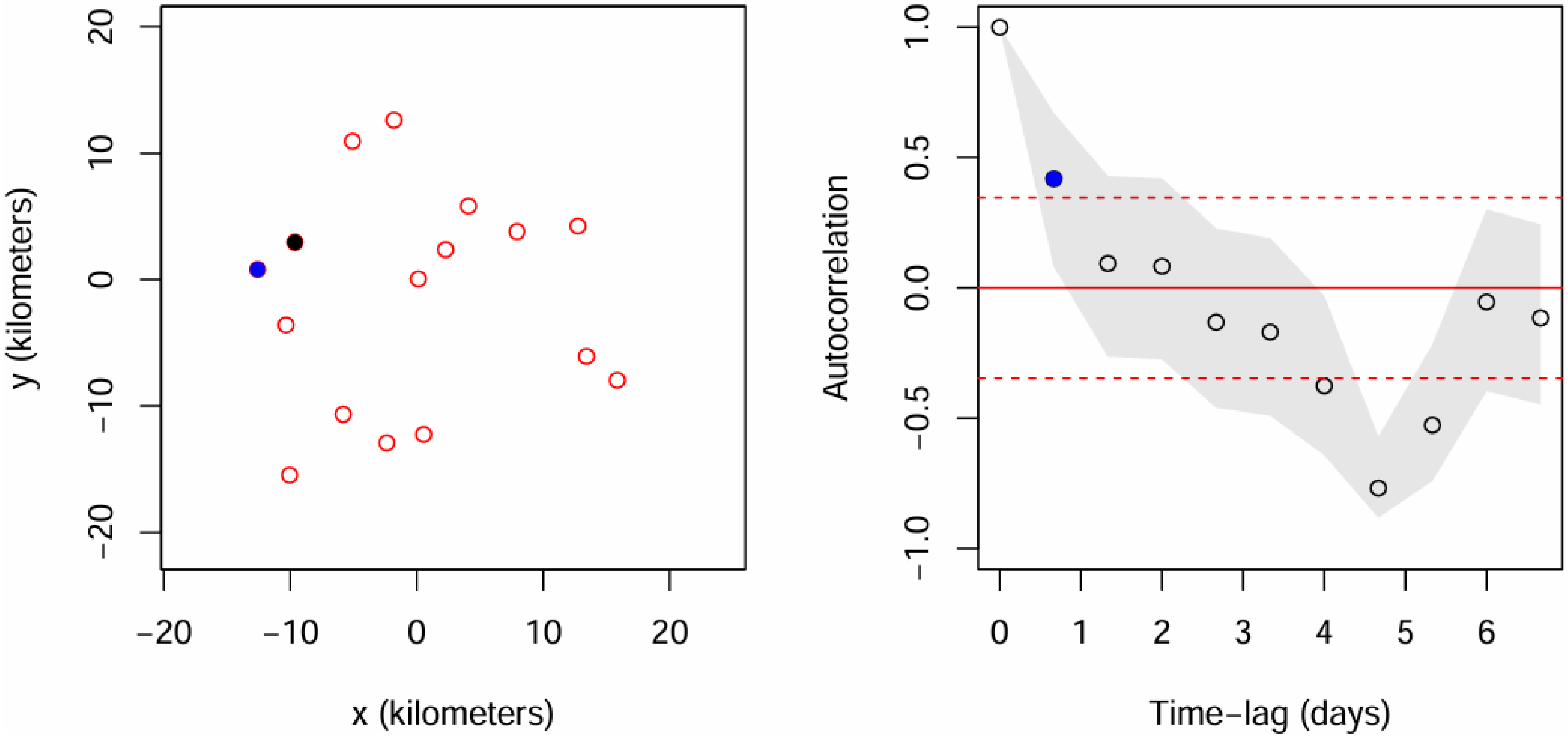

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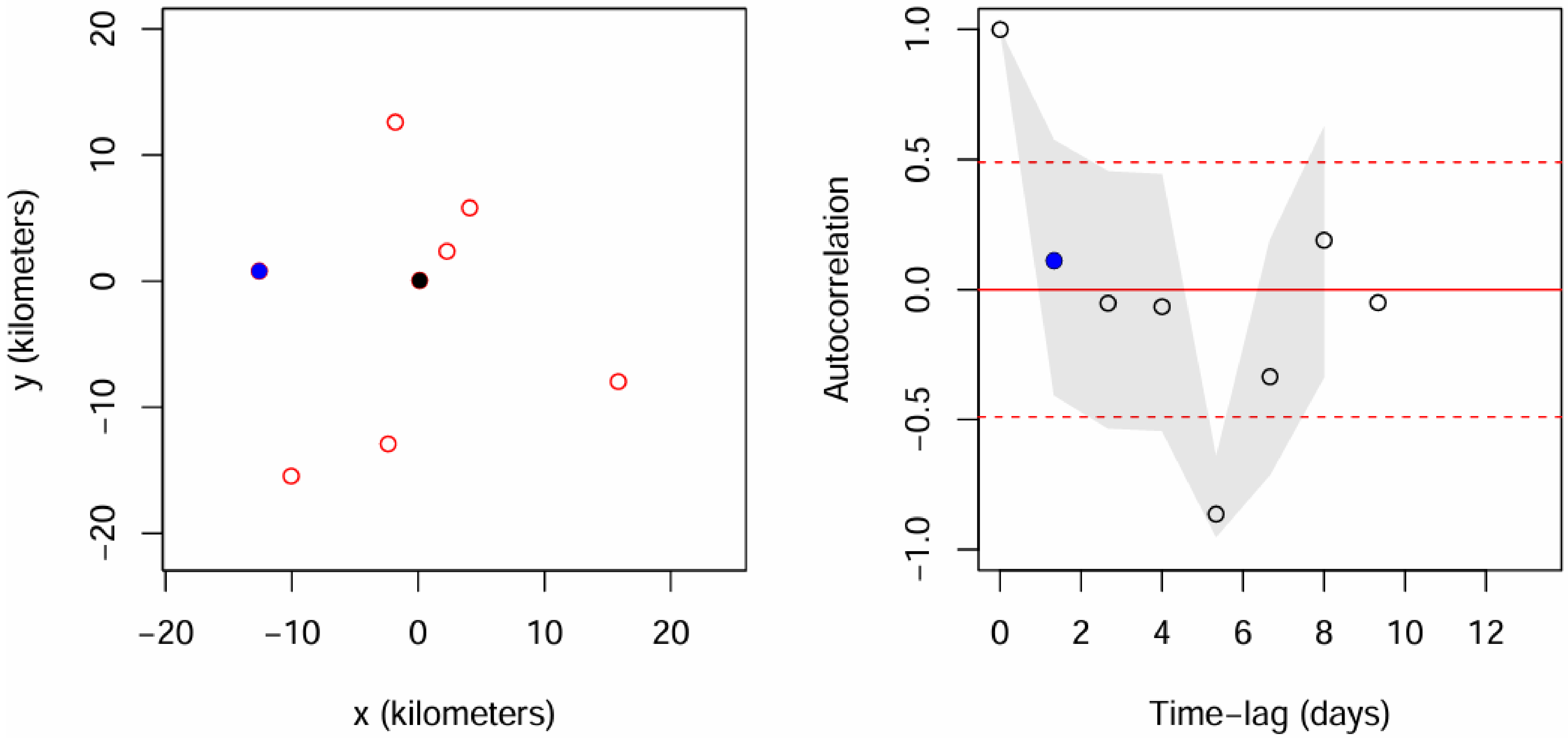

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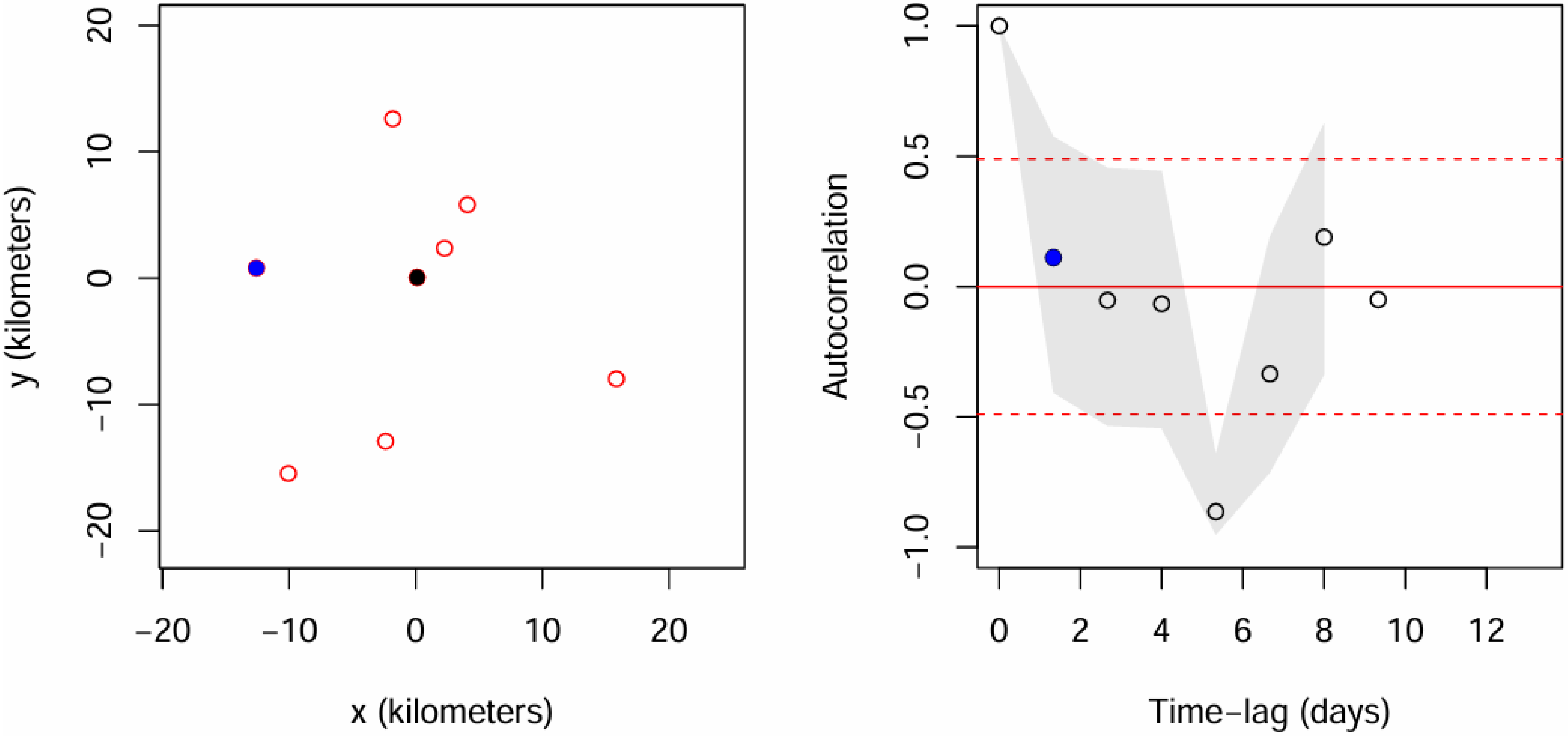
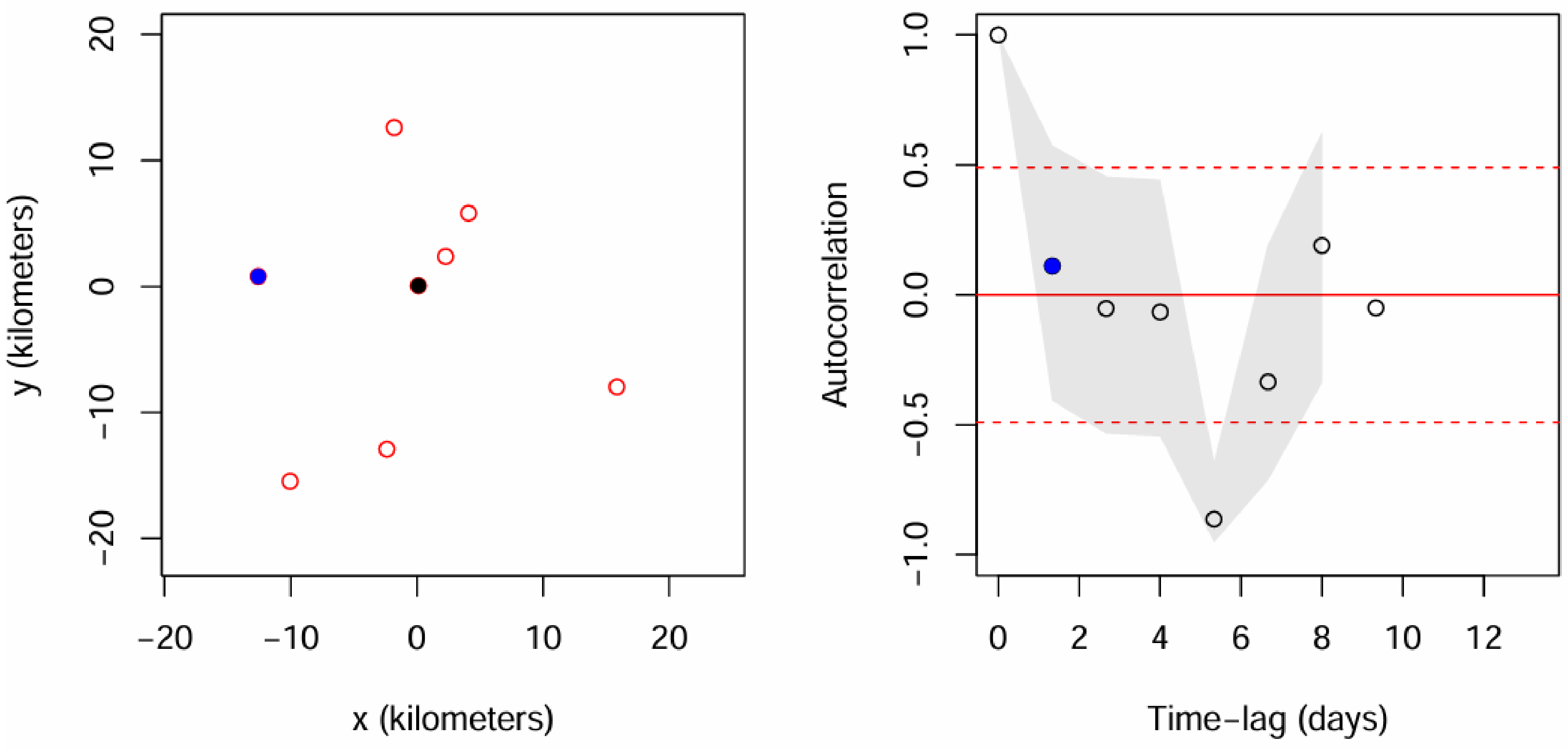
What about speed estimation with these data?



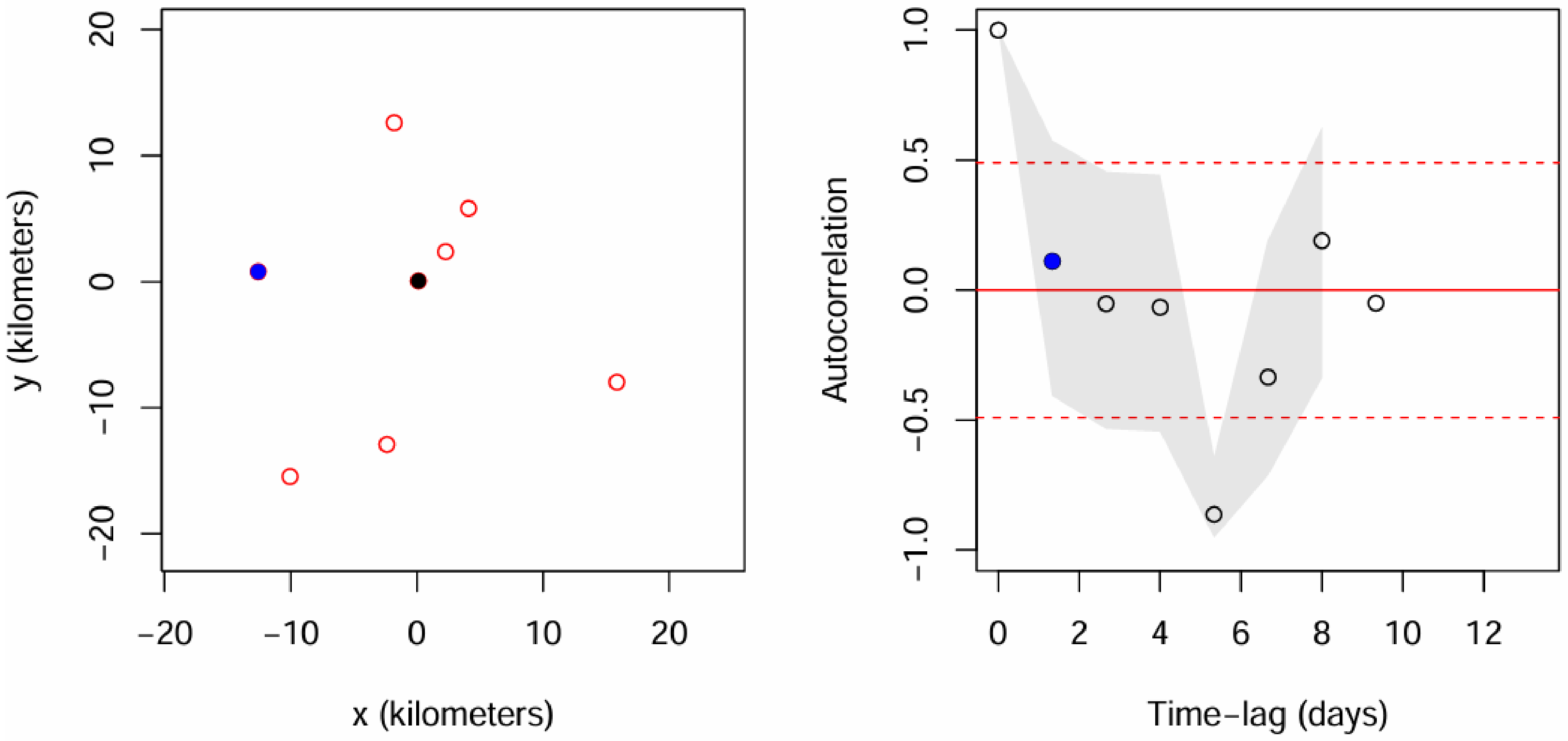
Autocorrelation is information

- *Convenient* for home-range analysis



Autocorrelation is information

- *Convenient* for home-range analysis
- *Worthless* for speed/distance estimation



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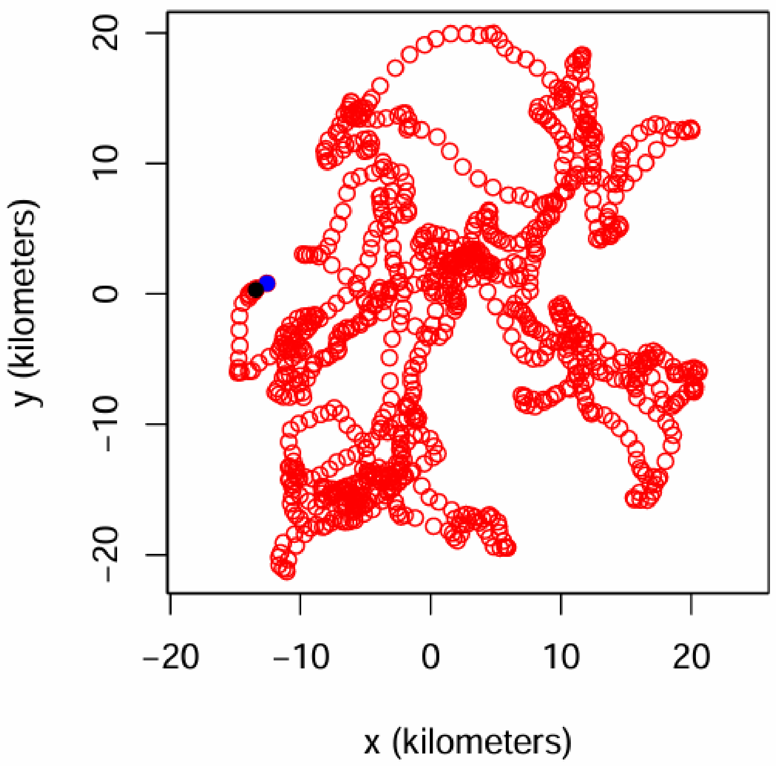
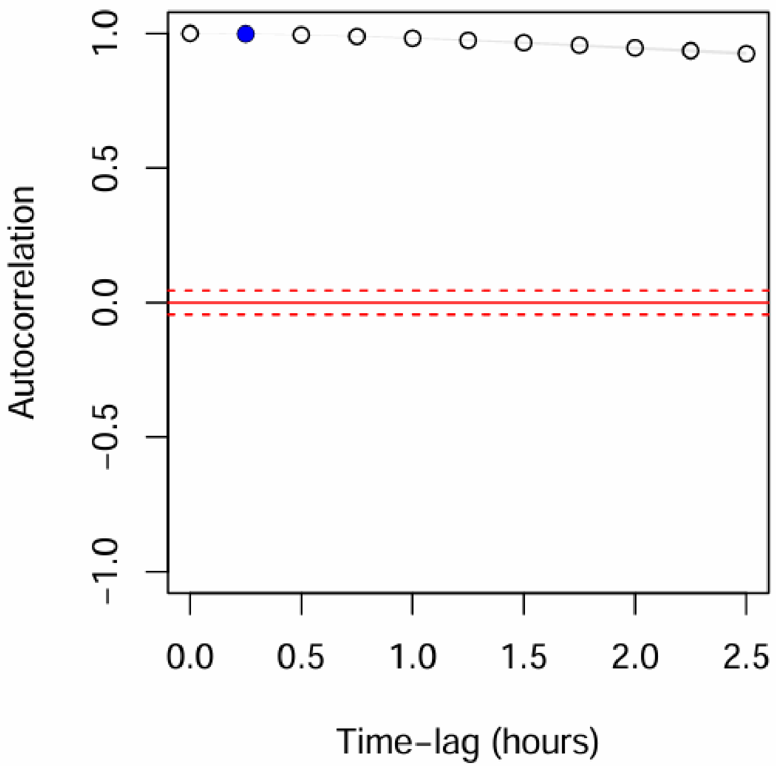
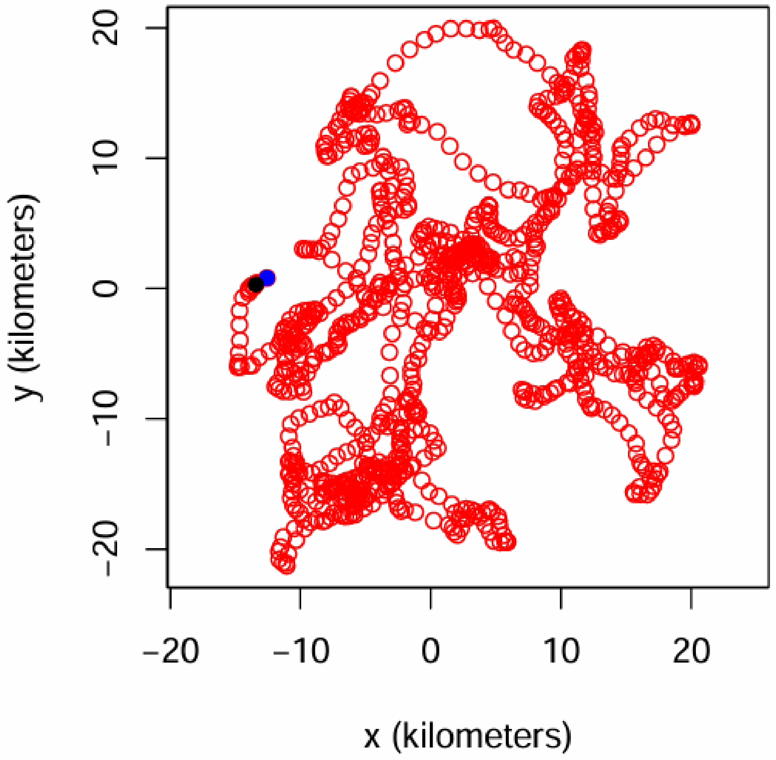
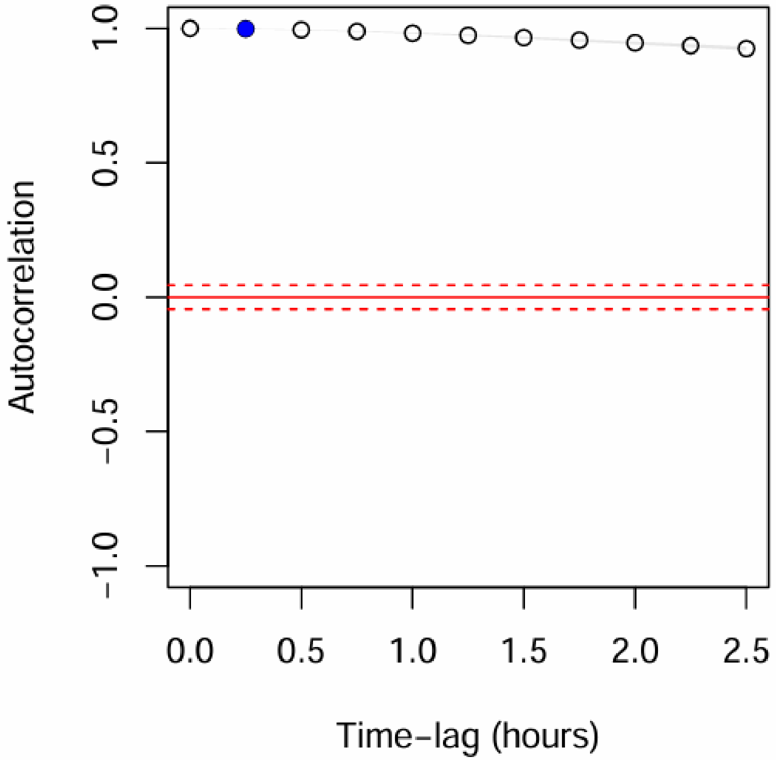




Autocorrelation is information

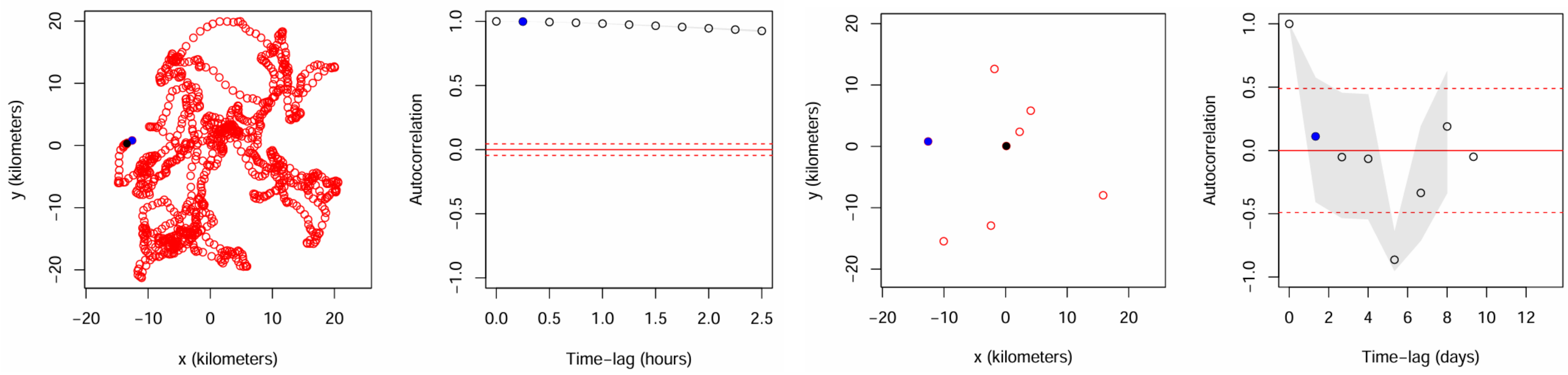
- *Inconvenient* for (conventional) home-range analysis
- *Great* for speed/distance estimation





Objective

We want methods that can handle whatever autocorrelation is present in the data



Overview

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2) The problem: Underestimating space use
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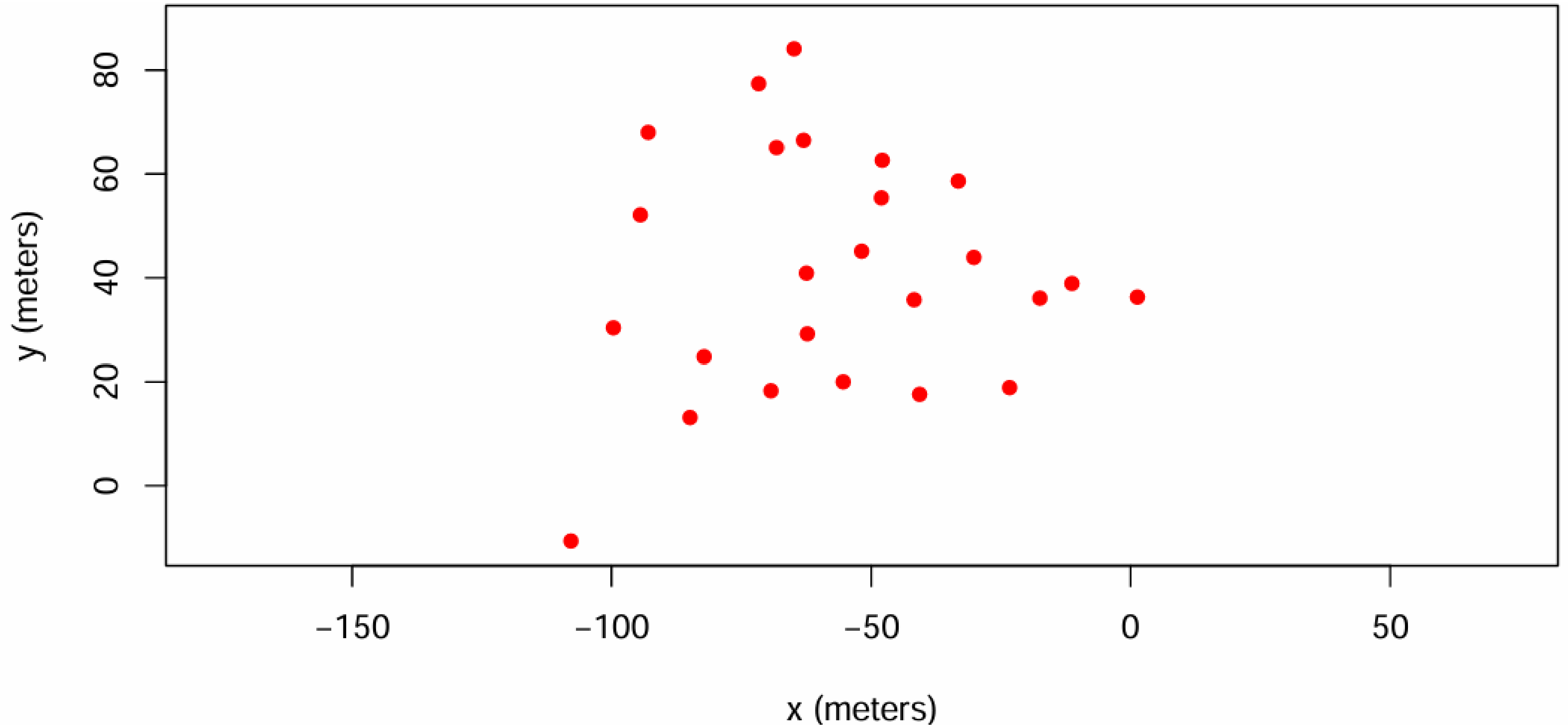
What do we mean by continuous time vs. discrete time?



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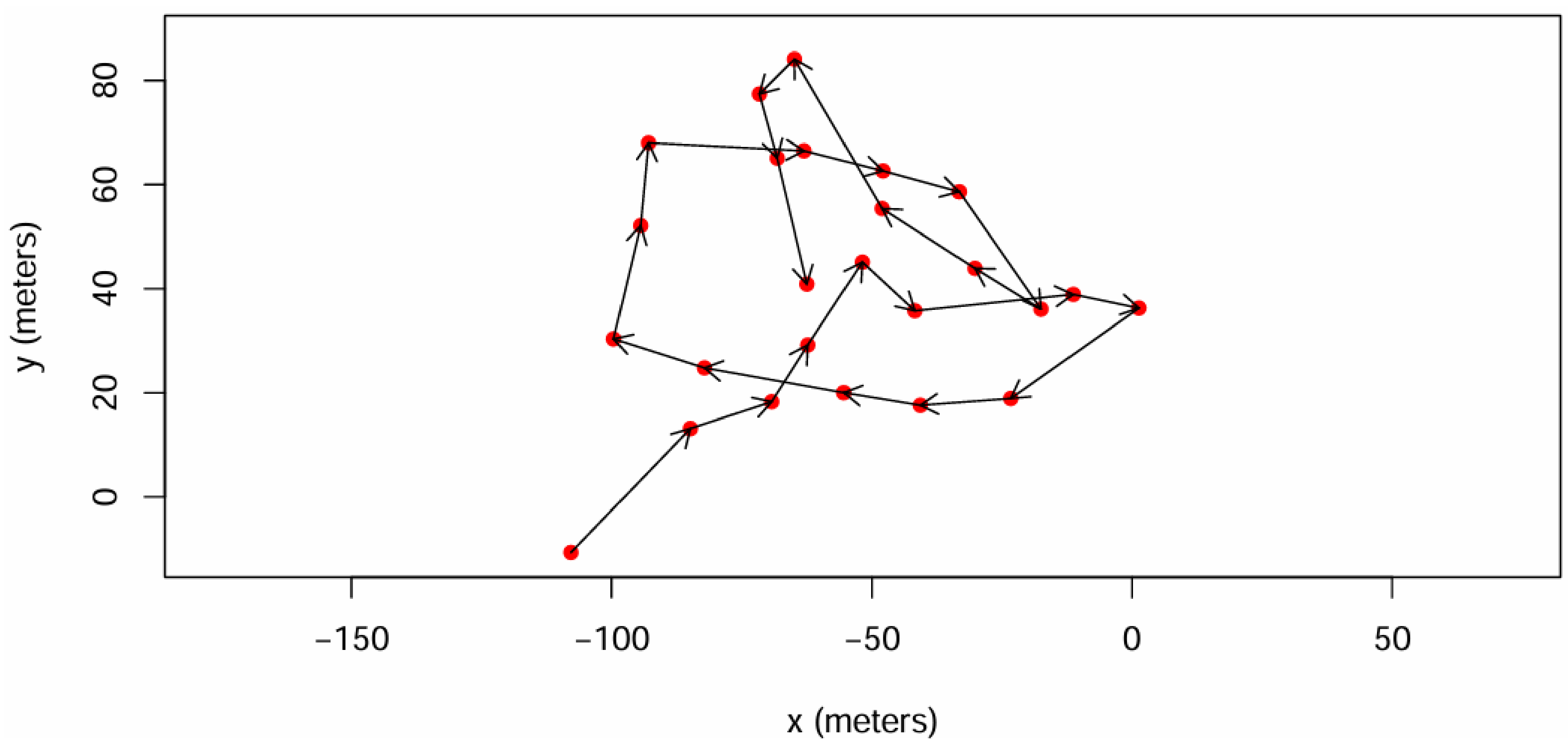

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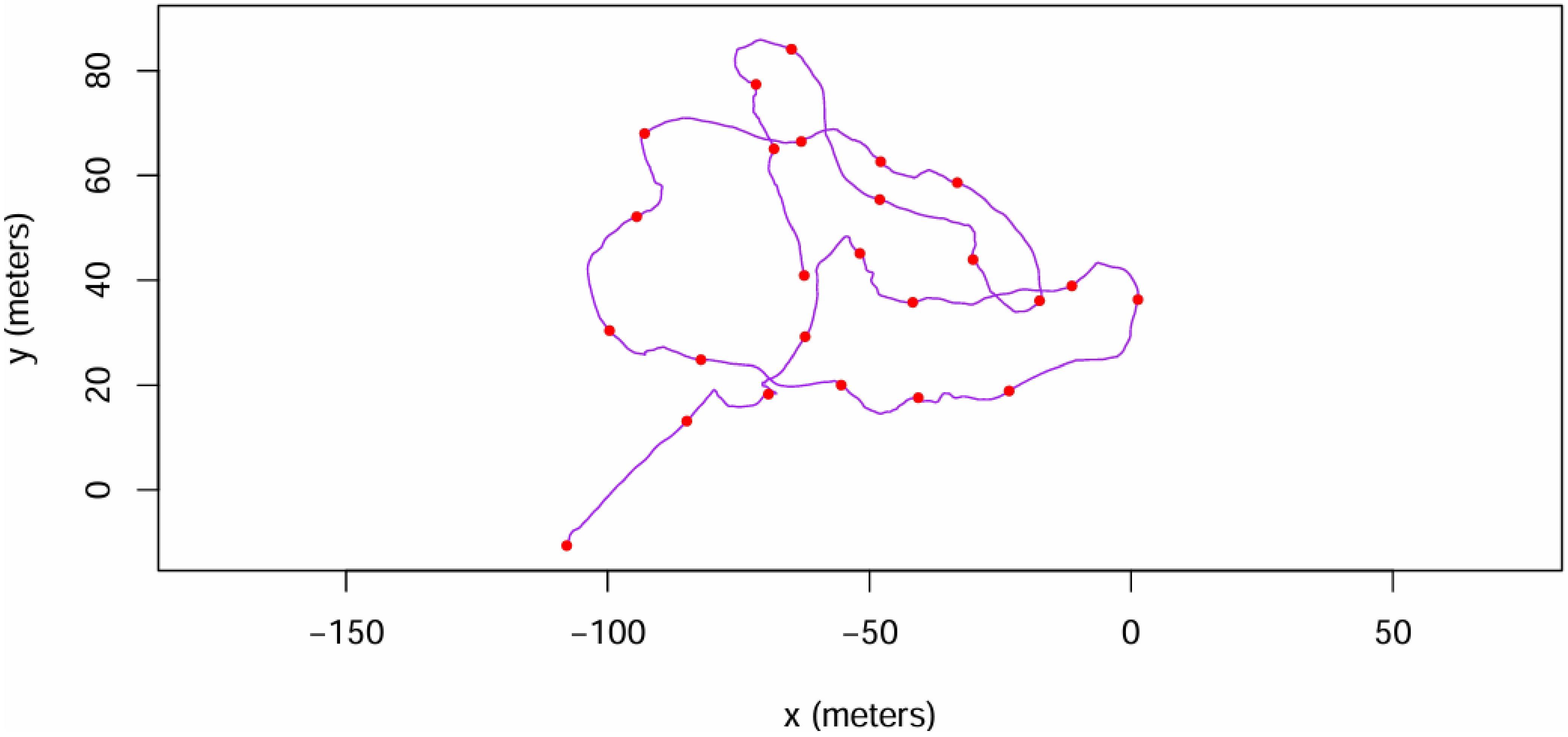

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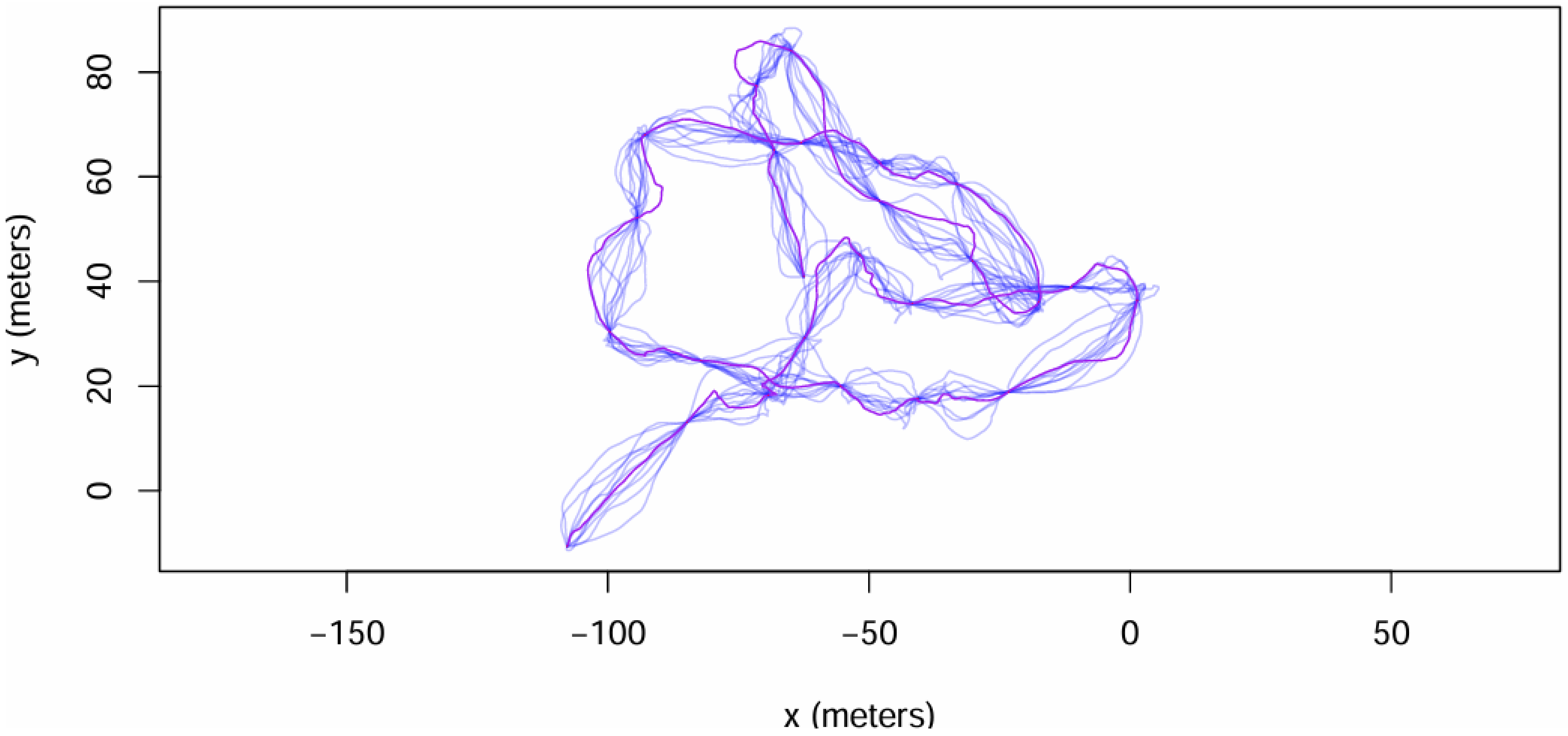

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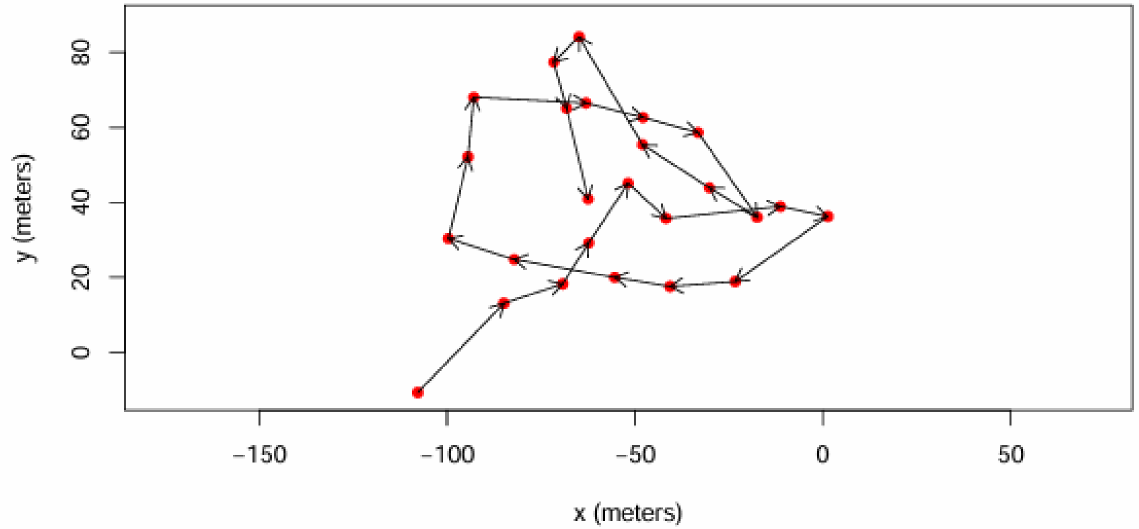
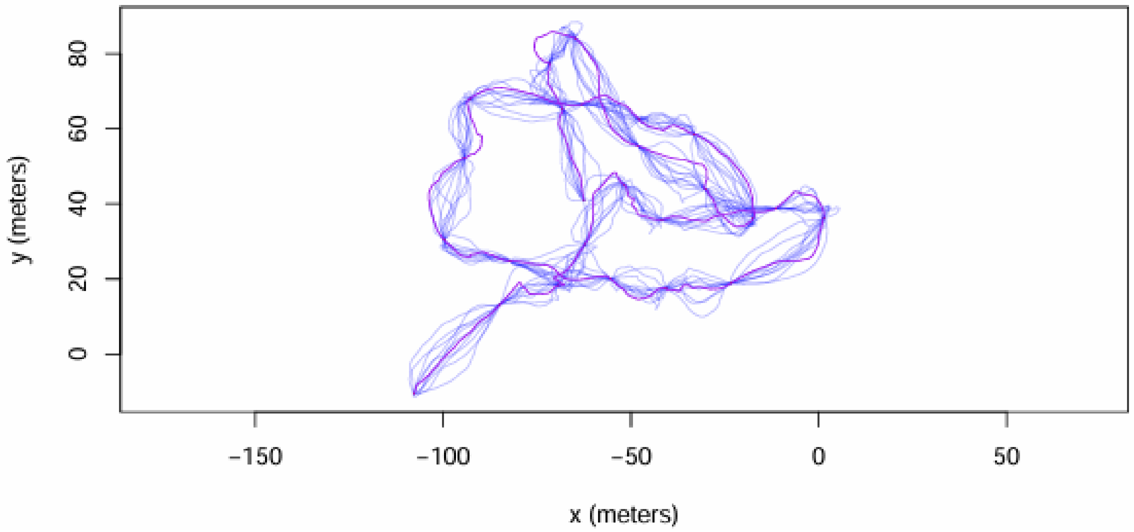
  - *Location error* is easy to model (vs. CRWs)

Motivating example:
Neglecting autocorrelation in speed estimation





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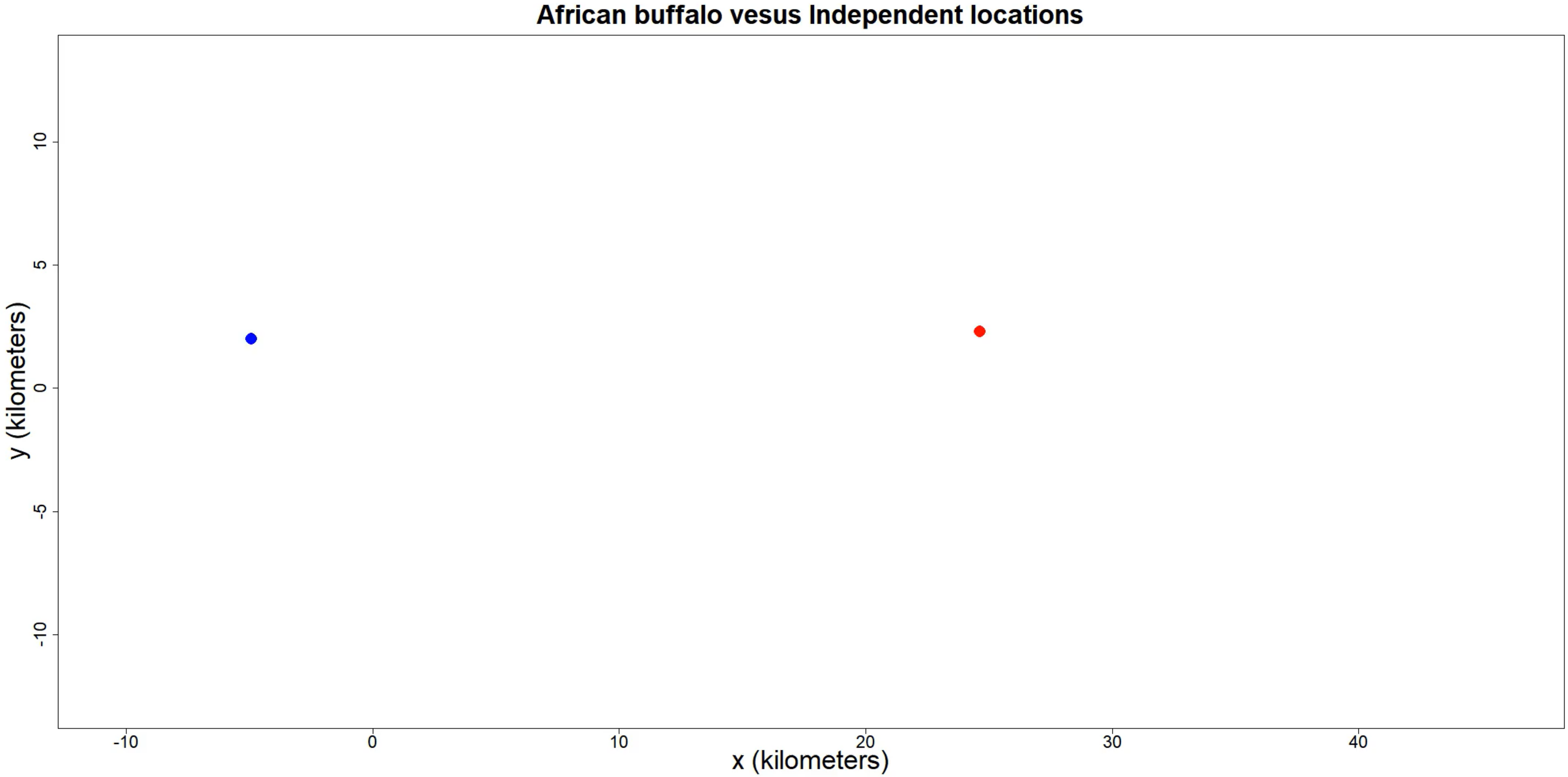
Accounting for non-independence: Stochastic process models

Building-block continuous-time stochastic process models

Accounting for non-independence: Stochastic process models

Building-block continuous-time stochastic process models

- Independent locations


African buffalo vesus Independent locations
y (kilometers)
10
5
0
-5
-10
-10
0
10
20
30
40
x (kilometers)

Accounting for non-independence: Stochastic process models

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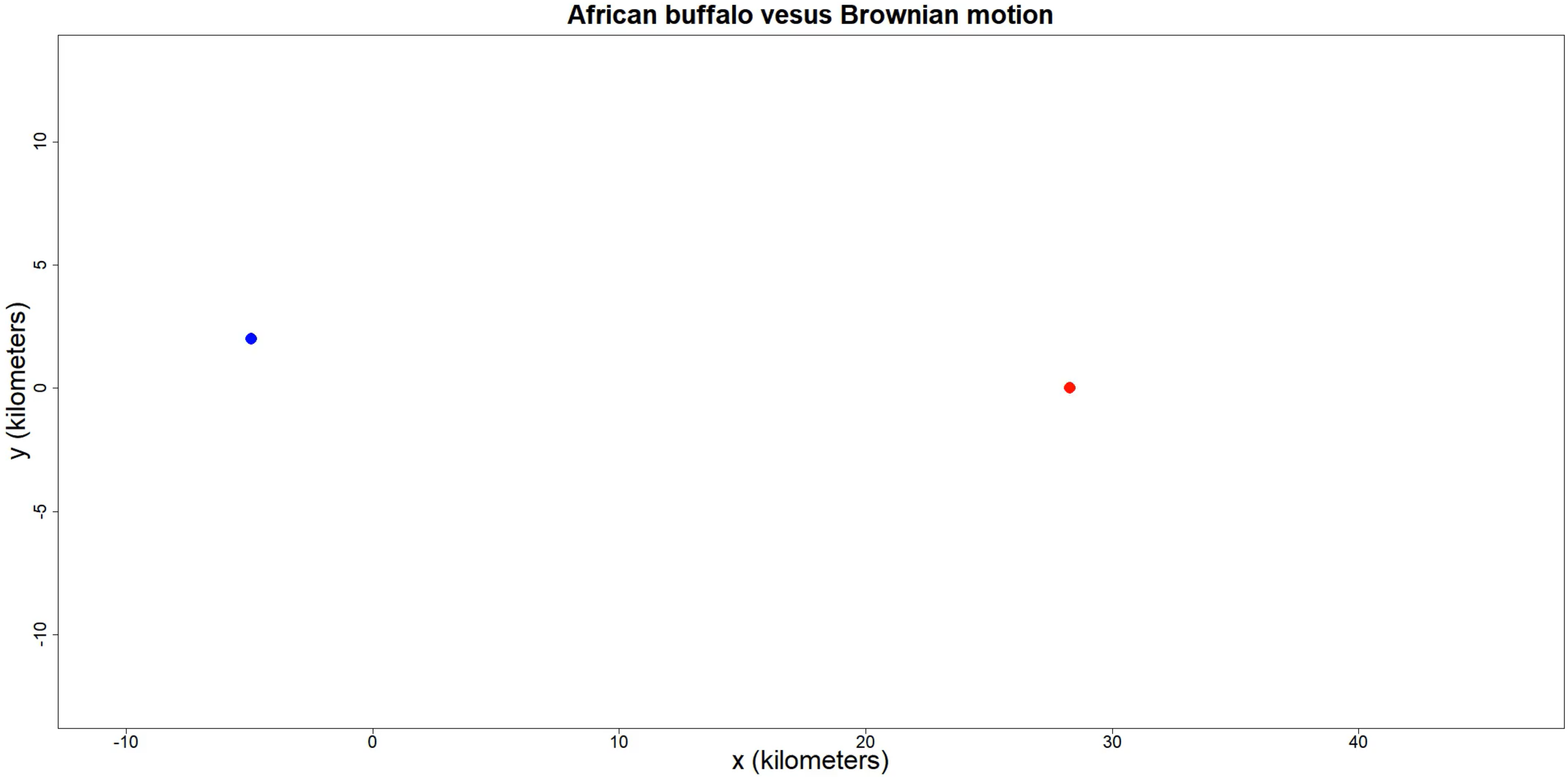
African buffalo vesus Brownian motion
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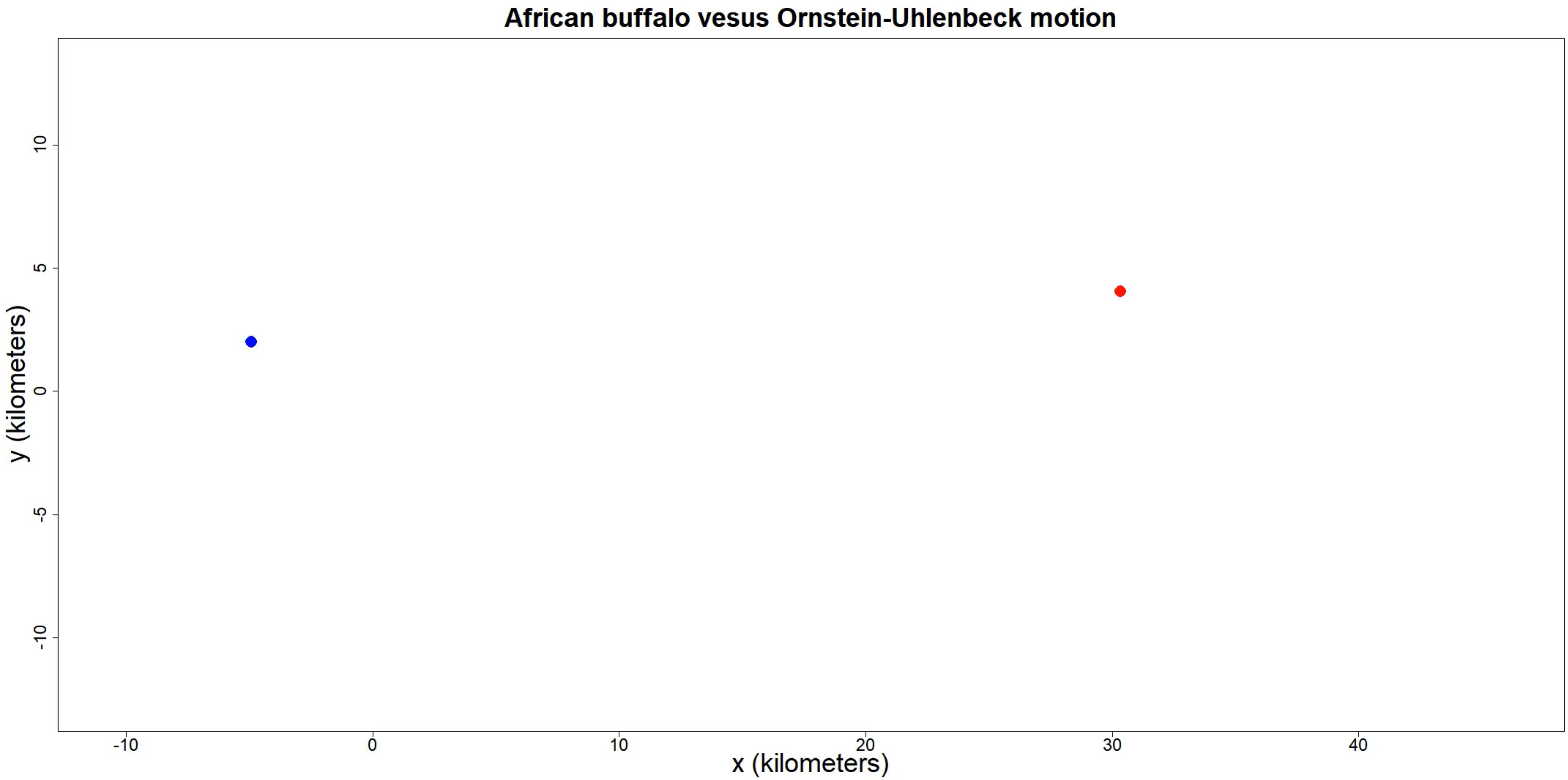
African buffalo vesus Ornstein-Uhlenbeck motion
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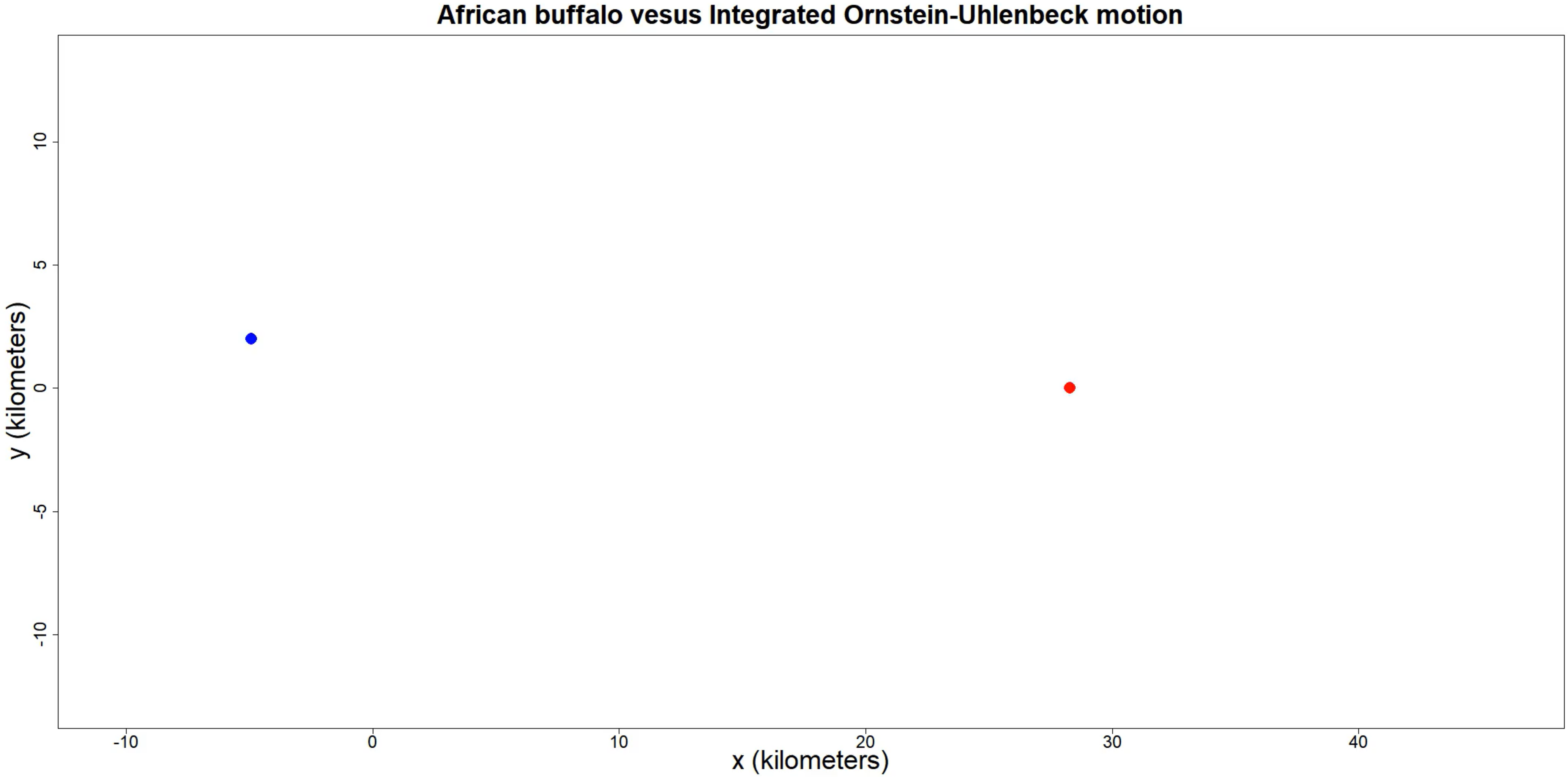
- Integrated Ornstein-Uhlenbeck motion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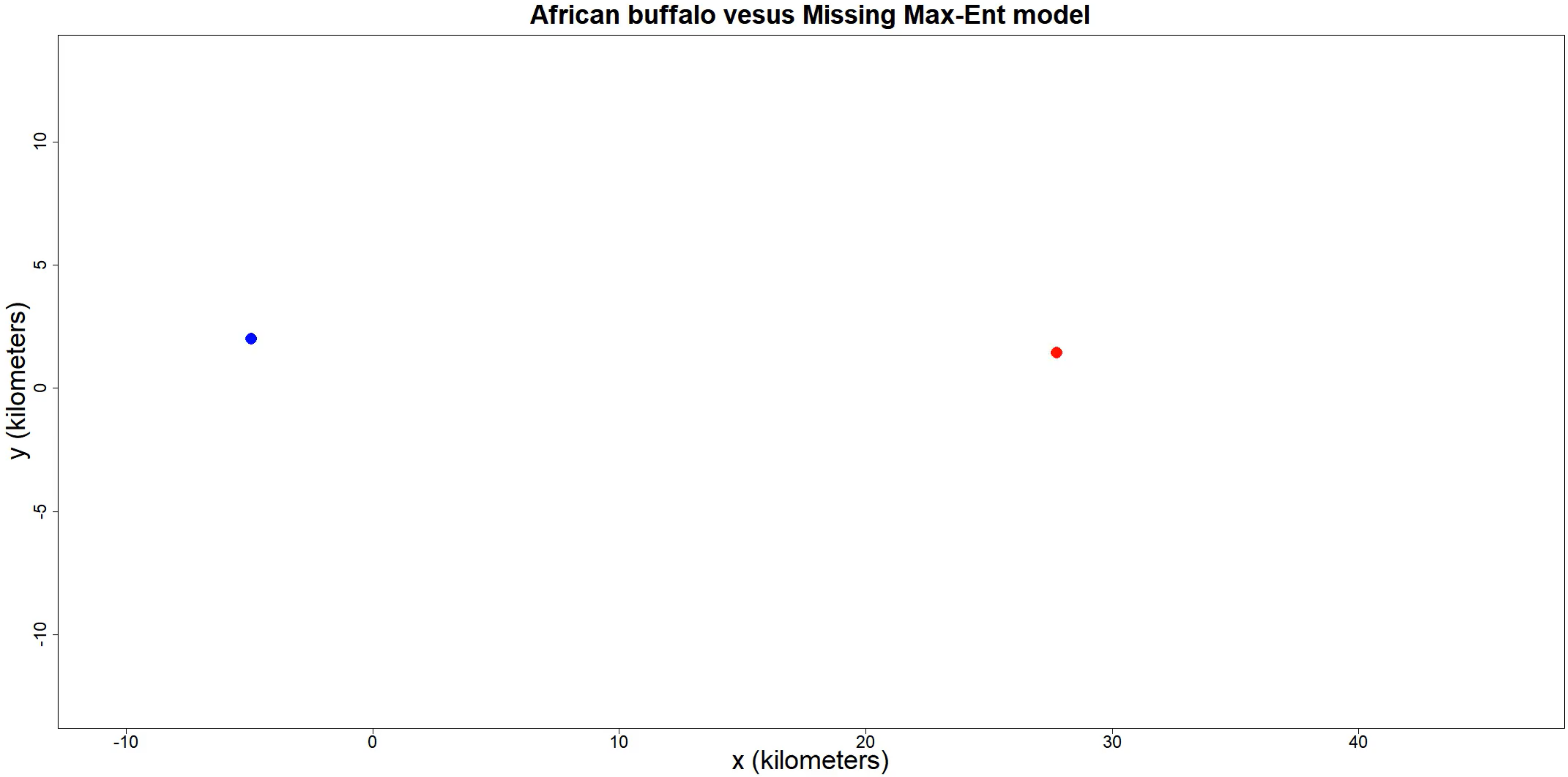
African buffalo vesus Missing Max-Ent model
y (kilometers)
10
5
0
-5
-10
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-10
0
10
20
30
40

Accounting for non-independence: Stochastic process models

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Accounting for non-independence: Stochastic process models

Building-block continuous-time stochastic process models

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Don't assume a model, ***select*** a model

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Let's dive into `R` `ctmmlearn` "ctmm_intro.R"

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ctmm_intro.R

Thank you

- Thanks to collaborators: Jesse Alston, Justin Calabrese, Bill Fagan, Michael J. Noonan, Roland Kays, Björn Reineking
- Funded by and supported by:

UCF
UNIVERSITY OF CENTRAL FLORIDA

NSF

Smithsonian
National Zoological Park
Conservation Biology Institute

CASUS
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UNIVERSITY OF MARYLAND 18 56